University of Zimbabwe

HMTH113 ORDINARY DIFFERENTIAL EQUATIONS

# The Theory Behind

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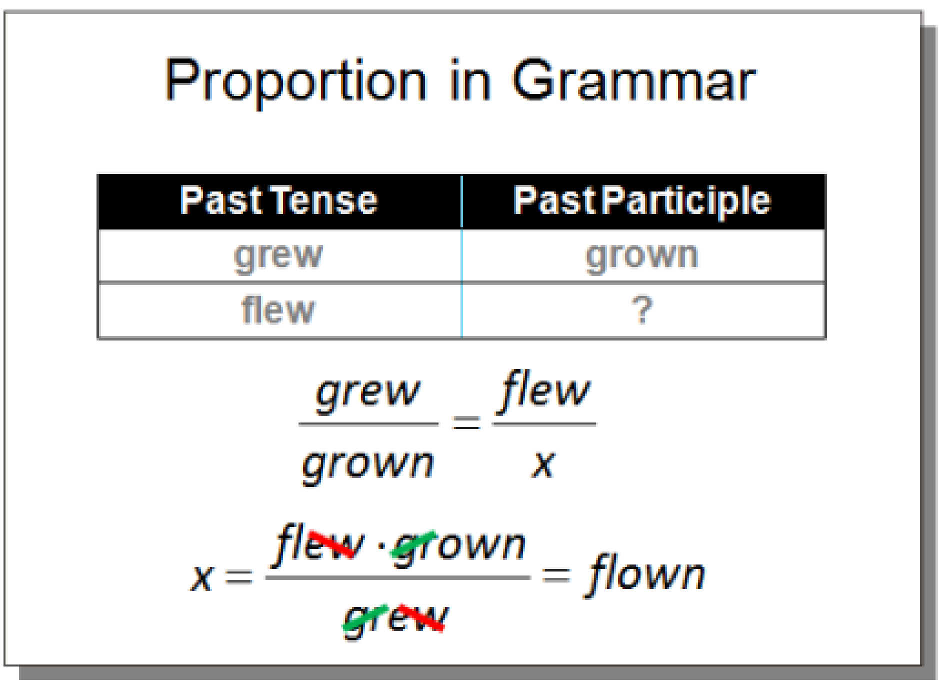


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## Abstract

*Everybody is a genius. But if you judge a fish by its ability to climb a tree, it will live its whole life believing that it is stupid.*

—Albert Einstein

The key point is that you are about to immerse yourself in serious Mathematics, with an emphasis on your attaining a deep understanding of the definitions, theorems, and proofs. Ordinary differential equations serve as mathematical models for many exciting “real-world” problems, not only in science and technology, but also in such diverse fields as economics, psychology, defense, and demography. Rapid growth in the theory of differential equations and in its applications to almost every branch of knowledge has resulted in a continued interest in its study by students in many disciplines. This has given ordinary differential equations a distinct place in mathematics curricula all over the world and it is now being taught at various levels in almost every institution of higher learning.

Thomas Jefferson (1743–1826), the author of the United States Declaration of Independence, wrote,

*If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of every one, and the receiver cannot dispossess himself of it. Its peculiar character, too, is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening me. That ideas should freely spread from one to another over the globe, for the moral and mutual instruction of man, and improvement of his condition, seems to have been peculiarly and benevolently designed by nature, when she made them, like fire, expansible over all space, without lessening their density in any point, and like the air in which we breathe, move, and have our physical being, incapable of confinement or exclusive appropriation.*

—Letter to Isaac McPherson, August 13, 1813

# Chapter 1

# Review and Collection of Formulas

*The laws of Nature are written in the language of mathematics. Math is a way to describe reality and figure out how the world works, a universal language that has become the gold standard of truth. In our world, increasingly driven by science and technology, mathematics is becoming, ever more, the source of power, wealth, and progress. Hence those who are fluent in this new language will be on the cutting edge of progress.*

— Edward Frenkel, Love and Math: The Heart of Hidden Reality

## 1.1 Limits and Continuity of Functions

In this chapter are some basic definitions and theorems from calculus most needed for differential equations.

**Definition 1.1.1.** *A function $f(x)$ has the* ***limit*** *$L$ as $x$ approaches $a$, in symbols*

$$\lim_{x\to a} f(x) = L,$$

*if, for every positive number $\epsilon$ there is a positive number $\delta$, such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.*

**Definition 1.1.2.** *A function $f(x)$ is defined to be* ***continuous at the point*** *$x = a$, if*

$$\lim_{x\to a} f(x) = f(a).$$

## 1.2 Derivatives and Integrals

The derivative of $f(x)$ is defined to be the limit

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = f'(x) = \frac{d}{dx}f(x).$$

Observe that if the derivative exists at $x$, then certainly $f(x)$ is continuous at $x$. The derivative of $y = f(x)$ is usually denoted by $y' = \frac{dy}{dx}$.However, in cases where the independent variable is the time $t$, Newton's notation $\dot{y}$ is often used.

A function may be continuous and yet fail to have a derivative.

**Example 1.2.1.** *$y = |x|$ does not have a derivative at $x = 0$.*

**Definition 1.2.1.** *A function $F(x)$ is an **indefinite integral** of $f(x)$,*

$$F(x) = \int f(x), dx,$$

*if*

$$\frac{d}{dx}F(x) = f(x).$$

# Chapter 2

# Introduction

*Any man who can drive safely while kissing a pretty girl is simply not giving the kiss the attention it deserves.*

—Albert Einstein

We live in a world of inter-related changing entities. The position of the earth changes with time, the velocity of a falling body changes with distance, the bending of a beam changes with the weight of the load placed on it, the area of a circle changes with the size of the radius, the path of a projectile changes with the velocity and angle at which it is fired.

In the language of mathematics, changing entities are called variables and the rate of change of one variable with respect to another a derivative. Equations which express a relationship among these variables and their derivatives are called differential equations. In both the natural and social sciences many of the problems with which they are concerned give rise to such differential equations. But what we are interested in knowing is not how the variables and their derivatives are related but only how the variables themselves are related. For example, from certain facts about the variable position of a particle and its rate of change with respect to time, we wish to determine how the position of the particle is related to the time so that we can know where the particle was, is, or will be at any time $t$. Differential equations thus originate whenever a universal law is expressed by means of variables and their derivatives.

An **ordinary differential equation** is a relation containing one real independent variable $x \in \mathbb{R} = (-\infty, \infty)$, the real dependent variable $y$, and some of its derivatives

$$y', y'', y''' \dots, y^{(n)} ({}' = \frac{d}{dx}).$$

The expressions $y', y'', y''', y^{(4)}, \cdots, y^{(n)}$ are often used to represent, respectively, the first, second, third, fourth, ... , $n$th derivatives of $y$ with respect to the independent variable under consideration.

Thus $y''$ represents $\dfrac{d^2y}{dx^2}$, if the independent variable is $x$, but represents $\dfrac{d^2y}{dp^2}$ if the independent variable is $p$. Observe that parenthesis are used in $y^{(n)}$ to distinguish it from the $n$th power, $y^n$.

For example,

$$\begin{aligned} xy' + 3y &= 6x^3 \\ y'^2 - 4y &= 0 \\ x^2y'' - 3xy' + 3y &= 0 \\ 2x^2y'' - y'^2 &= 0. \end{aligned}$$

The **order** of an ordinary differential equation is defined to be the order of the highest derivative in the equation. Thus, first two equations are first order, whereas the last two are second order.

In general, an *nth*-order differential equation can be written as

$$F(x, y, y', y'', \ldots, y^{(n)}) = 0,$$

where $F$ is a known function.

A functional relation between the dependent variable $y$ and the independent variable $x$, that, in some interval $J$, satisfies the given differential equation is said to be a *solution* of the equation.

**Example 2.0.2.** *The function* $y(x) = 7e^x + x^2 + 2x + 2$ *is a solution of the differential equation* $y' = y - x^2$ *in* $J = \mathbb{R}$.

The *general solution* of an *nth*-order differential equation depends on $n$ arbitrary constants, i.e., the solution $y$ depends on $x$ and the real constants $c_1, c_2, \ldots, c_n$.

**Example 2.0.3.** *The function* $y(x) = x^2 + ce^x$ *is the general solution of the differential equation* $y' = y - x^2 + 2x$ *in* $J = \mathbb{R}$.

The differential equation $xy' + 3y = 6x^3$ has the general solution $y(x) = x^3 + \dfrac{c}{x^3}$. The function $y(x) = x^3$ is a *particular solution* of the equation, and it can be obtained easily by taking the particular value of $c$ as 0 in the general solution.

For the differential equation $y'^2 - xy' + y = 0$, the general solution is $y(x) = cx - c^2$ and $y(x) = \dfrac{x^2}{4}$, is a particular solution not included in the general solution. This "extra" solution, which cannot be obtained by assigning particular values of the constants, is called a *singular solution*.

In the "general solution", the word *general* must not be taken in the sense of *complete*. A totality of all solutions of a differential equation is called a *complete solution*.

Consider the equation $y'''^2 - 2y'y''' + 3y''^3$. This is a third-order differential equation and we say that this is of degree 2, whereas, the second order differential equation $xy'' + 2y' + 3y - 6e^x = 0$ is of degree 1. In general, if a differential equation has the form of an *algebraic equation* of degree $k$ in the highest derivative, then we say that the given differential equation is of *degree* $k$.

We shall assume the the differential equation

$$F(x, y, y', y'', \ldots, y^{(n)}) = 0,$$

can be solved explicitly for $y^{(n)}$ in terms of the remaining $(n+1)$ quantities as

$$y^{(n)} = f(x, y, y', \ldots, y^{(n-1)}),$$

where $f$ is a known function.

**Example 2.0.4.** *The differential equation $y'^2 = 4y$ represents two differential equations, $y' = \pm 2\sqrt{y}$.*

Differential equations are classified into two groups : linear and non-linear. A differential equation is said to be **linear** if its linear in $y$ and all its derivatives. Thus, an *nth*-order linear differential equation has the form

$$\mathcal{P}_n[y] = p_0(x)y^{(n)} + p_1(x)y^{(n-1)} + \cdots + p_n(x)y = r(x).$$

**Example 2.0.5.** $xy' + 3y = 6x^3$ *and* $x^2y'' - 3xy' + 3y = 0$ *are linear and* $y'^2 - 4y = 0$ *and* $2x^2y'' - y'^2 = 0$ *are non-linear.*

If $r(x) = 0$, then it is called *homogeneous differential equation*, otherwise it is said to be a *non-homogeneous differential equation*.

A differential equation along with subsidiary conditions on the unknown function and its derivatives, all given at the same value of the independent variable, constitutes an **initial-value problem**. The subsidiary conditions are **initial conditions**. If the subsidiary conditions are given at more than one value of the independent variable, the problem is a **boundary-value problem** and the conditions are **boundary conditions**.

**Example 2.0.6.** *The problem $y'' + 2y' = e^x$, $y(\pi) = 1$, $y(\pi) = 2$ is an initial value problem, because the two subsidiary conditions are both given at $x = \pi$. The problem $y'' + 2y' = e^x$, $y(0) = 1$, $y(1) = 2$ is a boundary-value problem, because the two subsidiary conditions are given at $x = 0$ and $x = 1$.*

A solution to an initial-value or boundary-value problem is a function $y(x)$ that both solves the differential equation and satisfies all given subsidiary conditions.

# Chapter 3

# First Order Differential Equations

*Gravitation cannot be held responsible for people falling in love. How on earth can you explain in terms of chemistry and physics so important a biological phenomenon as first love? Put your hand on a stove for a minute and it seems like an hour. Sit with that special girl for an hour and it seems like a minute. That's relativity.*

—Albert Einstein

## 3.1 Standard and Differential Forms

Standard form for a first-order differential equation in the unknown function $y(x)$ is

$$y' = f(x, y)$$

or in **differential form**

$$M(x, y)dx + N(x, y)dy = 0$$

where $M(x, y)$ and $N(x, y)$ are functions. If $f(x, y)$ can be written as $f(x, y) = -p(x)y + q(x)$, the differential equation is **linear**. First-order linear differential equation can always be expressed as

$$y' + p(x)y = q(x).$$

## 3.2 Solutions of First Order Differential Equations

### 3.2.1 Separable Equations

The solution to the first-order separable differential equation

$$A(x)dx + B(y)dy = 0$$

is

$$\int A(x)dx + \int B(y)dy = c$$

where $c$ is an arbitrary constant.

**Example 3.2.1.** *Solve* $\dfrac{dy}{dx} = \dfrac{x^2+2}{y}$.

***Solution:*** *Can be written in the differential form*

$$(x^2+2)dx - ydy = 0$$

*which is separable with* $A(x) = x^2 + 2$ *and* $B(y) = -y$. *Its solution is*

$$\int (x^2+2)dx - \int ydy = c$$

*or*

$$\frac{1}{3}x^3 + 2x - \frac{1}{2}y^2 = c.$$

**Example 3.2.2.** *Solve* $(1+x)dy - ydx = 0$.

***Solution:*** *Dividing by* $(1+x)y$, *we can write* $\dfrac{dy}{y} = \dfrac{dx}{1+x}$, *from which it follows that*

$$\begin{aligned}
\int \frac{dy}{y} &= \int \frac{dx}{1+x} \\
\ln|y| &= \ln|1+x| + c_1 \\
y &= e^{\ln|1+x|+c_1} \\
y &= e^{c_1}(1+x) = c(1+x).
\end{aligned}$$

**Example 3.2.3.** *Solve the initial value problem* $\dfrac{dy}{dx} = -\dfrac{x}{y}$, $y(4) = -3$.

***Solution:*** *Rewriting the equation as* $ydy = -xdx$, *we get*

$$\int ydy = -\int xdx \quad \textit{or} \quad \frac{y^2}{2} = -\frac{x^2}{2} + c.$$

*Now when* $x = 4$, $y = -3$, *so we have* $x^2 + y^2 = 25$.

**Example 3.2.4.** *Solve $y' = -2xy$, $y(0) = 1.8$.*

***Solition:*** *By separation and integration, $\frac{dy}{y} = -2xdx$, $\ln y = -x^2 + d$ or $y = ce^{-x^2}$. This is the general solution. From it and the initial condition $y(0) = ce^0 = c = 1.8$. Hence the IVP has the solution $y = 1.8e^{-x^2}$.*

## 3.3 Use of Substitutions : Homogeneous Equations

If a function $f$ possess the property

$$f(tx, ty) = t^\alpha f(x, y)$$

for some real number $\alpha$, then $f$ is said to be a *homogeneous function* of degree $\alpha$.

**Example 3.3.1.** *The function $f(x, y) = x^3 + y^3$ is homogeneous of degree 3 because*

$$f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y).$$

A differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is said to be *homogeneous* if both coefficients $M$ and $N$ are homogeneous functions of the **same** degree. Such an equation can be reduced to separable variables by either the substitution $y = ux$ or $x = vy$, where $u$ and $v$ are new dependent variables.

**Example 3.3.2.** *Solve $(x^2 + y^2)dx + (x^2 - xy)dy = 0$.*

***Solution:*** *Inspection of $M(x, y) = x^2 + y^2$ and $N(x, y) = x^2 - xy$ shows that these coefficients are homogeneous functions of degree 2. If we let $y = ux$, then*

$$dy = udx + xdu,$$

*so that, after substituting, the given equation becomes*

$$\begin{aligned}
(x^2 + u^2x^2)dx + (x^2 - ux^2)[udx + xdu] &= 0 \\
x^2(1 + u)dx + x^3(1 - u)du &= 0 \\
\frac{1 - u}{1 + u}du + \frac{dx}{x} &= 0 \\
\left[-1 + \frac{2}{1 + u}\right] du + \frac{dx}{x} &= 0.
\end{aligned}$$

*After integrating, we have*

$$\begin{aligned}
-u + 2\ln|1 + u| + \ln|x| &= \ln|c| \\
-\frac{y}{x} + 2\ln\left|1 + \frac{y}{x}\right| + \ln|x| &= \ln|c|.
\end{aligned}$$

*Using the properties of logarithms, we can write the preceding solution as*

$$\ln\left|\frac{(x+y)^2}{cx}\right| = \frac{y}{x}.$$

*The definition of a logarithm then yields* $(x+y)^2 = cxe^{\frac{y}{x}}$.

## 3.4 Exact Equations

A differential equation

$$M(x,y)dx + N(x,y)dy = 0 \tag{3.1}$$

is **exact** if there exists a function $g(x,y)$ such that

$$dg(x,y) = M(x,y)dx + N(x,y)dy.$$

### Test for Exactness

If $M(x,y)$ and $N(x,y)$ are continuous functions and have continuous first partial derivatives, then Equation (3.1) is **exact** if and only if

$$\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}.$$

**Example 3.4.1.** *Solve* $2xydx + (x^2-1)dy = 0$.

***Solution:*** *With* $M(x,y) = 2xy$ *and* $N(x,y) = x^2-1$, *we have* $\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}$. *Thus the equation is exact and so there exists a function* $g(x,y)$ *such that* $\frac{\partial g}{\partial x} = 2xy$ *and* $\frac{\partial g}{\partial y} = x^2-1$. *From the first of the equations we obtain, after integrating* $g(x,y) = x^2y + h(y)$. *Taking the partial derivative of the last expression with respect to* $y$ *and setting the result equal to* $N(x,y)$, *gives*

$$\frac{\partial g}{\partial y} = x^2 + h'(y) = x^2 - 1.$$

*It follows* $h'(y) = -1$ *and* $h(y) = -y$. *Hence* $g(x,y) = x^2y - y$, *so the solution of the equation is*

$$x^2y - y = c.$$

**Example 3.4.2.** *Determine whether the differential equation*

$$(x + \sin y)dx + (x\cos y - 2y)dy = 0$$

*is exact. If it is, solve it.*

***Solution:*** *Here* $M(x,y) = x + \sin y$ *and* $N(x,y) = x\cos y - 2y$. *Thus* $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} = \cos y$, *and the differential equation is exact.*

*We seek a function* $g(x,y)$ *such that* $\dfrac{\partial g}{\partial x} = x + \sin y$. *Integrating both sides of this equation with respect to* $x$, *we find*

$$\begin{aligned} \int \frac{\partial g}{\partial x} dx &= \int (x + \sin y) dy \\ g(x,y) &= \frac{1}{2}x^2 + x\sin y + h(y). \end{aligned} \tag{3.2}$$

*Differentiating Equation* (3.2) *with respect to* $y$, *yields* $\dfrac{\partial g}{\partial y} = x\cos y + h'(y)$ *and we find that*

$$x\cos y + h'(y) = x\cos y - 2y \quad \textit{or} \quad h'(y) = -2y$$

*from which it follows that* $h(y) = -y^2 + c_1$. *Substituting, we have*

$$g(x,y) = \frac{1}{2}x^2 + x\sin y - y^2 + c_1.$$

*The solution of the differential equation is*

$$\frac{1}{2}x^2 + x\sin y - y^2 = c_2 \quad \textit{where} \quad c_2 = c - c_1.$$

### 3.4.1 Linear Equations

A first-order linear differential equation has the form

$$y' + p(x)y = q(x). \tag{3.3}$$

An integrating factor for Equation (3.3) is

$$I(x) = e^{\int p(x)dx}. \tag{3.4}$$

**Example 3.4.3.** *Solve* $y' + \left(\dfrac{4}{x}\right) y = x^4$.

***Solution:*** *The differential equation has the form of Equation* (3.3), *with* $p(x) = \dfrac{4}{x}$ *and* $q(x) = x^4$ *and is linear. Here*

$$\int p(x)dx = \int \frac{4}{x} dx = 4\ln x = \ln x^4,$$

*so Equation* (3.4) *becomes*

$$I(x) = e^{\int p(x)dx} = e^{\ln x^4} = x^4. \tag{3.5}$$

*Multiplying the differential equation by the integrating factor defined by* (3.5), *we obtain*

$$x^4 y' + 4x^3 y = x^8 \quad \text{or} \quad \frac{d}{dx}(x^4 y) = x^8.$$

*Integrating both sides with respect to $x$ of the last equation, we obtain*

$$x^4 y = \frac{1}{9}x^9 + c \quad \text{or} \quad y = \frac{c}{x^4} + \frac{x^5}{9}.$$

**Example 3.4.4.** *Solve* $x\dfrac{dy}{dx} - 4y = x^6 e^x$.

***Solution:*** *Dividing by $x$, we get the standard form*

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x.$$

*From this we identify,* $p(x) = -\dfrac{4}{x}$ *and* $p(x) = x^5 e^x$. *Hence the integrating factor is*

$$e^{-4\int \frac{dx}{x}} = e^{-4\ln x} = e^{\ln x^{-4}} = x^{-4}.$$

*Multiplying the differential equation by $x^{-4}$ and rewrite*

$$x^{-4}\frac{dy}{dx} - 4x^{-4}y = xe^x \quad \text{or} \quad \frac{d}{dx}(x^{-4}y) = xe^x.$$

*Integrating by parts we have*

$$x^{-4}y = xe^x - e^x + c \quad \text{or} \quad y = x^5 e^x - x^4 e^x + cx^4.$$

**Example 3.4.5.** *Solve* $\dfrac{dy}{dx} + y = x$, $y(0) = 4$.

***Solution:*** *The equation is in standard form and $p(x) = 1$ and $q(x) = x$. The integrating factor* $e^{\int dx} = e^x$, *so integrating*

$$\frac{d}{dx}(e^x y) = xe^x$$

*gives*

$$e^x y = xe^x - e^x + c \quad \text{or} \quad y = x - 1 + ce^{-x}.$$

*From the initial condition we know that $y = 4$ when $x = 0$, hence $c = 5$. Hence the solution is*

$$y = x - 1 + 5e^{-x}.$$

**Example 3.4.6.** *Find the general solution of the following differential equation*

$$x\frac{dy}{dx} + (1 + x)y = e^x.$$

***Solution:*** *This is a linear differential equation of first order*

$$\frac{dy}{dx} + \frac{1+x}{x}y = \frac{e^x}{x}.$$

*The integrating factor is given by*

$$e^{\int \frac{1+x}{x}dx}.$$

*Since* $\int \frac{1+x}{x}dx = \int \left(\frac{1}{x}+1\right)dx = \ln x + x$, *therefore integrating factor is* $e^{\ln x + x} = xe^x$. *Multiplying the differential equation with the integrating factor, we have* $\frac{d}{dx}[xe^x y] = e^{2x}$. *Integrating we have* $xye^x = \frac{e^{2x}}{2} + c$. *Therefore* $y = \frac{e^{2x}}{2x} + \frac{ce^{-x}}{x}$.

## 3.5 The Bernoulli Equation

**Definition 3.5.1.** *A **Bernoulli equation** is the equation of the form*

$$\frac{dy}{dx} + a(x)y = f(x)y^n, \quad n \neq 0, 1. \tag{3.6}$$

When $n = 0, 1$ the Bernoulli equation becomes a linear equation.

Set $z = y^{1-n}$. Then $z' = (1-n)y^{-n}y'$. So if we multiply both sides of equation (3.6) by $(1-n)y^{-n}$, we obtain

$$(1-n)y^{-n}y' + (1-n)a(x)y^{1-n} = (1-n)f(x)$$

or

$$\frac{dz}{dx} + (1-n)a(x)z = (1-n)f(x).$$

This equation is now linear and can be solved as before.

**Example 3.5.1.** *Solve*

$$\frac{dy}{dx} - \frac{y}{x} = -\frac{5}{2}x^2y^3.$$

***Solution:*** *Here* $n = 3$ *so let* $z = y^{-2}$. *Let* $z' = -2y^{-3}y'$ *and multiply both sides of the equation by* $-2y^{-3}$ *to obtain*

$$-2y^{-3}y' + \frac{2}{x}y^{-2} = 5x^2$$

*or*

$$z' + \frac{2z}{x} = 5x^2. \tag{3.7}$$

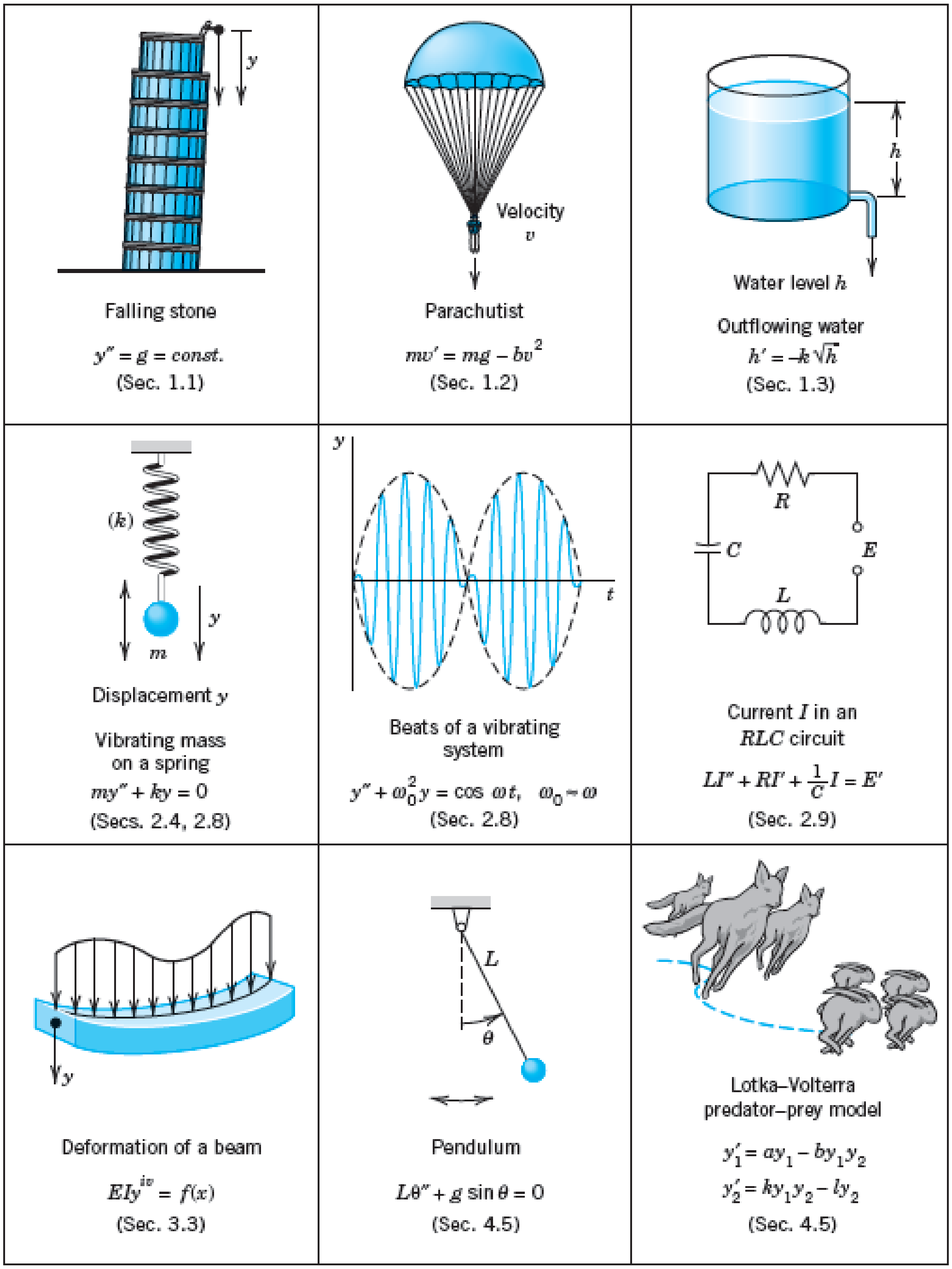
*The integrating factor for this linear equation is* $e^{2\int \frac{dx}{x}} = e^{2\ln x} = e^{\ln x^2} = x^2$. *Multiplying both sides of equation* (3.7) *by* $x^2$, *we have* $x^2z' + 2xz = 5x^4$ *or* $(x^2z)' = 5x^4$. *Thus*

$$x^2z = 5\int x^4dx + c = x^5 + c.$$

*Hence*

$$y^{-2} = z = x^3 + cx^{-2} \quad or \quad y = (x^3 + cx^{-2})^{-\frac{1}{2}}.$$

## 3.6 Applications of First Order Differential Equations



### 3.6.1 Growth and Decay Problems

Let $N(t)$ denote the amount of substance (or population) that is either growing or decaying. If we assume that $\frac{dN}{dt}$, the time rate of change of this amount of substance, is proportional to the amount of substance present, then

$$\frac{dN}{dt} = kN \quad \text{or} \quad \frac{dN}{dt} - kN = 0,$$

where $k$ is the constant of proportionality.

### 3.6.2 Temperature Problems

Newton's law of cooling, which is equally applicable to heating, states that *the time rate of change of the temperature of a body is proportional to the temperature difference between the body and its surrounding medium.* Let $T$ denote the temperature of the body and let $T_m$ denote the temperature of the surrounding medium. Then the time rate of change of the temperature of the body is $\frac{dT}{dt}$, and Newton's law of cooling can be formulated as $\frac{dT}{dt} = -k(T - T_m)$, or as

$$\frac{dT}{dt} + kT = kT_m,$$

where $k$ is a positive constant of proportionality.

**Example 3.6.1.** *A body cools in air of constant temperature $T_m = 20°C$. If the temperature of the body changes from $100°C$ to $60°C$ in $20minutes$, determine how much more time it will need for the temperature to fall to $30°C$.*

***Solution:*** *Newton's Law of Cooling requires that*

$$\frac{dT}{dt} = -k(T - T_m).$$

*The general solution is*

$$\int \frac{dT}{T - T_m} = -k \int dt + C \Rightarrow \ln|T - T_m| = -kt + \ln C \qquad \therefore T = T_m + Ce^{-kt}.$$

*At $t = 0$, $T = 100°C$, then*

$$100 = 20 + Ce^{-k\cdot 0} = 20 + C \qquad \Rightarrow C = 80.$$

*At $t = 20min$, $T = 60°C$, then*

$$60 = 20 + 80e^{-k\cdot 20} \qquad \Rightarrow k = -\frac{1}{20}\ln\left(\frac{60-20}{80}\right) = 0.03466.$$

*Hence*

$$T = 20 + 80e^{-0.03466t} \qquad or \qquad t = -\frac{1}{0.03466}\ln\left(\frac{T-20}{80}\right) mins.$$

*When* $T = 30°C$,

$$t = -\frac{1}{0.03466}\ln\left(\frac{30-20}{80}\right) = 60mins.$$

*Hence, it will need another* $60 - 20 = 40$*minutes for the temperature to fall to* $30°C$.

### 3.6.3 Falling Body Problems

Consider a vertically falling body of mass $m$ that is being influenced only by gravity $g$ and an air resistance that is proportional to the velocity of the body. Assume that both gravity and mass remain constant and, for convenience, choose the downward direction as the positive direction. For the problem at hand, there are two forces acting on the body: the force due to gravity given by the weight $w$ of the body, which equals $mg$, and the force due to air resistance given by $kv$, where $k \geq 0$ is a constant of proportionality. The minus sign is required because this force opposes the velocity; that is, it acts in the upward, or negative, direction. The net force $F$ on the body is, therefore, $F = mg - kv$. Substituting in Newton's Second law, we obtain $mg - kv = m\frac{dv}{dt}$ or

$$\frac{dv}{dt} + \frac{k}{m}v = g,$$

as the equation of motion for the body.

### 3.6.4 Electrical Circuits

The basic equation governing the amount of current $I$ (in amperes) in a simple $RL$ circuit consisting of a resistance $R$ (in ohms), an inductor $L$ (in henries), and an electromotive force (abbreviated emf) $E$ (in volts) is

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E}{L}.$$

For an $RC$ circuit consisting of a resistance, a capacitance $C$ (in farads), an emf, and no inductance, the equation governing the amount of electrical charge $q$ (in coulombs) on the capacitor is

$$\frac{dq}{dt} + \frac{1}{RC}q = \frac{E}{R}.$$

## 3.7 Problems

1. Find the general solution of the following, (i) $\frac{dy}{dx} = \frac{y-1}{xy}$ (ii) $xydy = (y-1)(x+1)dx$ (iii) $L\frac{di}{dt} + Ri = 0$ (iv) $\frac{dx}{dy} = \frac{1+2y^2}{y\sin x}$ (v) $(x^2+1)\sin y dy + 2x\cos y dx = 0$

(vi) $x^3+y^3\dfrac{dy}{dx}=0$ (vii) $y'=1-y^2$ (viii) $2\sqrt{y}e^{2x}dx+\dfrac{dy}{1+y}=0$ (ix) $e^{x-y}dy+dx=0$
(x) $xy'=4y$.

2. Determine whether the given equation is exact. If it is exact, solve.
(i) $x\dfrac{dy}{dx}=2xe^x-y+6x^2$ (ii) $(y\ln y-e^{-xy})dx+\left(\dfrac{1}{y}+x\ln y\right)dy=0$
(iii) $(x^3+y^3)dx+3xy^2dy=0$ (iv) $\left(x^2y^3-\dfrac{1}{1+9x^2}\right)\dfrac{dy}{dx}+x^3y^2=0$.

3. Determine the most general function $M(x,y)$ such that the following equation is exact

$$M(x,y)dx+(2ye^x+y^2e^{3x})dy=0.$$

4. Determine a function $M(x,y)$ so the following differential equation is exact

$$M(x,y)dx+\left(xe^{xy}+2xy+\frac{1}{x}\right)dy=0.$$

5. Determine a function $N(x,y)$ so that the following differential equation is exact

$$\left(y^{\frac{1}{2}}x^{-\frac{1}{2}}+\frac{x}{x^2+y}\right)dx+N(x,y)dy=0.$$

6. Solve the following differential equation by using an appropriate substitution.
(i) $(x-y)dx+xdy=0$ (ii) $(y^2+yx)dx-x^2dy=0$ (iii) $2xydx-(x^2+y^2)dy=0$
(iv) $xy^2dy=(x^3+y^3)dx$ (v) $ydx=2(x+y)dy$ (vi) $\dfrac{dy}{dx}=\dfrac{x+3y}{3x+y}$ (vii) $\dfrac{dy}{dx}=\dfrac{y-x}{y+x}$.

7. Solve the following initial value problems.
(i) $(3x^2y+8xy^2)+(x^3+8x^2y+12y^2)y'=0$, $y(2)=1$ (ii) $(x-y\cos x)-\sin xy'=0$, $y(\frac{\pi}{2})=1$
(iii) $(4x^3e^{x+y}+x^4e^{x+y}+2x)+(x^4e^{x+y}+2y)y'=0$, $y(0)=1$.

8. Solve the given differential equation.
(i) $x^2y'+x(x+2)y=e^x$ (ii) $x\dfrac{dy}{dx}+(3x+1)y=e^{-3x}$ (iii) $xy'+2y=e^x+\ln x$
(iv) $x^2-1)\dfrac{dy}{dx}+2y=(x+1)^2$ (v) $\dfrac{dr}{d\theta}+r\sec\theta=\cos\theta$ (vi) $\cos x\dfrac{dy}{dx}+y\sin x=1$.

9. Solve these Bernoulli equations.
(i) $\dfrac{dy}{dx}-y=xy^5$ (ii) $\dfrac{dy}{dx}+2xy+xy^4=0$ (iii) $\dfrac{dy}{dx}+y=y^2(\cos x-\sin x)$
(iv) $\dfrac{dy}{dx}+\dfrac{1}{3}y=\dfrac{1}{3}(1-2x)y^4$ (v) $\dfrac{dy}{dx}=y(xy^3-1)$ (vi) $x\dfrac{dy}{dx}-(1+x)y=xy^2$.

10. The number of supermarkets $C(t)$ throughout the country that are using a computerised checkout system is described by the initial value problem

$$\frac{dC}{dt}=C(1-0.0005C),\ C(0)=1,$$

where $t>0$. How many supermarkets are using the computerized method when $t=10$? How many companies are estimated to adopt the new procedure over a period of time?

11. The population $P(t)$ at any time in a suburb of a large city is governed by the initial value problem

$$\frac{dP}{dt} = P(10^{-1} - 10^{-7}P), \quad P(0) = 5000,$$

where $t$ is measured in months. What is the limiting value of the population? At what time will be population be equal to one-half of this limiting value?

12. Obtain a solution of the equation

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X)$$

governing second order reactions in the two cases $\alpha \neq \beta$ and $\alpha = \beta$.

13. In a third-order chemical reaction the number of grams $X$ of a compound obtained by combining three chemicals is governed by

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X)(\gamma - X).$$

Solve the equation under the assumption $\alpha \neq \beta \neq \gamma$.

# Chapter 4

# Higher Order Differential Equations

*Go to Heaven for the climate, Hell for the company.*

—Mark Twain

## 4.1 Introduction

We turn now to the solution of ordinary differential equations of order two or higher. In the previous Chapter we saw that we could solve a few first-order differential equations by recognizing them as separable, linear, exact, homogeneous, or perhaps Bernoulli equations.

## 4.2 Initial Value and Boundary Value Problems

For a linear differential equation an $n$th-order initial-value problem is

$$\begin{aligned} \textit{Solve} \quad &: \quad a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \\ \textit{Subject to} \quad &: \quad y(x_0) = y_0,\, y'(x_0) = y_1, \ldots, y^{(n-1)}(x_0) = y_{n-1} \end{aligned}$$

where $y_0, y_1, \ldots, y_{n-1}$ are arbitrary constants, is called an *initial value problem*(IVP). The specified values of the unknown function and its first $n-1$ derivatives at a single point $x_0 : y_0, y'(x_0) = y_1, \ldots, y^{(n-1)}(x_0) = y_{n-1}$ are called *initial conditions.* We seek a solution on some interval $J$ containing $x_0$. In the case of a linear second order equation, a solution satisfying the initial conditions $y(x_0) = y_0$ and $y'(x_0) = y_1$ is a function satisfying the differential equation on $J$ whose graph passes through $(x_0, y_0)$ such that the slope of the curve at the point is the number $y_1$.

### 4.2.1 Existence of a Unique Solution

**Theorem 4.2.1.** *Let $a_n(x), a_{n-1}(x), \ldots, a_1(x), a_0(x)$ and $g(x)$ be continuous on an interval $J$ and let $a_n(x) \neq 0$ for every $x$ in this interval. If $x = x_0$ is any point in this interval, then a solution $y(x)$ of the initial value problem exists on this interval and is unique.*

**Example 4.2.1.** *The function $y(x) = 3e^{2x} + e^{-2x} - 3x$ is a solution of the initial value problem*

$$y'' - 4y = 12x, \quad y(0) = 4, \quad y'(0) = 1.$$

*Now the differential equation is linear, the coefficients as well as $g(x) = 12x$ are continuous, and $a_2(x) = 1 \neq 0$ on any interval containing $x = 0$. We conclude from Theorem that the given function is the unique solution.*

The requirements from the Theorem that $a_i(x), i = 0, 1, 2, \ldots, n$ be continuous and $a_n(x) \neq 0$ for every $x \in J$ are both important. Specifically, if $a_n(x) = 0$ for some $x$ in the interval, then the solution of a linear initial value problem may not be unique or even exist.

**Example 4.2.2.** *The function $y(x) = cx^2 + x + 3$ is a solution of the initial value problem*

$$x^2y'' - 2xy' + 2y = 6, \quad y(0) = 3, \quad y'(0) = 1$$

*on the interval $(-\infty, \infty)$ for any choice of the parameter $c$. In other words, there is no unique solution of the problem, since $a_2(x) = x^2$ is zero at $x = 0$ and the initial conditions are also imposed at $x = 0$.*

### 4.2.2 Boundary Value Problem

Another type of problem consists of solving a linear differential equation of order two or greater in which the dependent variable $y$ or its derivatives are specified at *different points*. A problem such as

$$\begin{aligned} \textit{Solve} &: \; a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \\ \textit{Subject to} &: \; y(a) = y_0, \quad y(b) = y_1 \end{aligned}$$

is called a **boundary value problem**(BVP). The prescribed values $y(a) = y_0$ and $y(b) = y_1$ are called **boundary conditions**. For a second order differential equation, other pairs of boundary conditions could be

$$\begin{aligned} y'(a) = y_0, &\quad y(b) = y_1 \\ y(a) = y_0, &\quad y'(b) = y_1 \\ y'(a) = y_0, &\quad y'(b) = y_1 \end{aligned}$$

where $y_0$ and $y_1$ denote **arbitrary constants**. These three pairs of conditions are just special cases of the general boundary conditions

$$\begin{aligned} \alpha_1 y(a) + \beta_1 y'(a) &= \gamma_1 \\ \alpha_2 y(b) + \beta_2 y'(b) &= \gamma_2. \end{aligned}$$

In general in all conditions of the Theorem are fulfilled, a boundary value problem may have
(i) several solutions (ii) a unique solution, or (iii) no solution at all.

The most general second order linear equation can be written as

$$y''(x) + a(x)y'(x) + b(x)y(x) = f(x) \tag{4.1}$$

where $a(x), b(x)$ and $f(x)$ are functions of the independent variable $x$ only.

If the function $f(x)$ is identically zero, we say equation (4.1) is **homogeneous**, otherwise it is **non-homogeneous**.

**Example 4.2.3.** *(a) $y'' + 2xy' + 3y = 0$ is homogeneous.*

*(b) $y'' + 2xy' + 3y = e^x$ is non-homogeneous.*

If the coefficient functions $a(x)$ and $b(x)$ are constants, $a(x) = a$ and $b(x) = b$, the equation is said to have **constant coefficients**. If either $a(x)$ or $b(x)$ is not constant, the equation is said to have **variable coefficients**.

**Example 4.2.4.** *(a) $y'' + 3y' - 10y = 0$ has constant coefficients.*

*(b) $y'' + 3xy' + 10x^2y = 0$ has variable coefficients.*

## 4.3 Linear Combination and Linear Independence

**Definition 4.3.1.** *Let $y_1$ and $y_2$ be any two functions. By a* ***linear combination*** *of $y_1$ and $y_2$ we mean a function $y(x)$ that can be written in the form*

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

*for some constants $c_1$ and $c_2$.*

**Definition 4.3.2.** *Two functions are* ***linearly independent*** *whenever the relation*

$$c_1y_1(x) + c_2y_2(x) = 0,$$

*for all $x$, implies that $c_1 = c_2 = 0$. Otherwise they are* ***linearly dependent****.*

**Example 4.3.1.** *Verify that the functions $y_1 = 1$ and $y_2 = x$ are linearly independent.*

*To determine linear independence, we consider*

$$c_1y_1 + c_2y_2 = c_1 \cdot 1 + c_2 \cdot x = 0.$$

*If $x = 0$, we have $c_1 \cdot 1 + c_2 \cdot 0 = c_1 = 0$. If $x = 1$ we get $c_2 = 0$. Hence $c_1 = c_2 = 0$. This proves that $y_1 = 1$ and $y_2 = x$ are linearly independent.*

If you have a collection of functions $y_1(x), y_2(x), \cdots, y_n(x)$. A linear combination of these functions is any function of the form

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$$

where $c_1, c_2, \cdots, c_n$ are constants. We shall say that the collection of functions $y_1(x), y_2(x), \cdots, y_n(x)$ are linearly independent if the equation

$$c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) n = 0,$$

when $c_1 = c_2 = \cdots = c_n = 0$.

**Theorem 4.3.1.** *Every homogeneous linear second order differential equation*

$$y'' + a(x)y' + b(x)y = 0$$

*has two linearly independent solutions.*

**Theorem 4.3.2.** *Let $y_1(x)$ and $y_2(x)$ be any two solutions of the homogeneous equation*

$$y'' + a(x)y' + b(x)y = 0.$$

*Then, any linear combination of them is also a solution of the differential equation.*

## 4.4 Superposition Principle : Homogeneous Equations

In the next theorem we see that the sum, or superposition, of two or more solutions of a homogeneous linear differential equation is also a solution.

**Theorem 4.4.1.** *Let $y_1, y_2, \ldots, y_k$ be solutions of the homogeneous linear nth-order differential equation on an interval $J$. Then the linear combination*

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x),$$

*where the $c_1, i = 1, 2, \ldots, k$ are arbitrary constants, is a solution on the interval.*

*Proof.* We prove the case when $n = k = 2$. Let $y_1(x)$ and $y_2(x)$ be solutions of

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0.$$

If we define $y = c_1 y_1(x) + c_2 y_2(x)$, then

$$\begin{aligned} a_2(x)[c_1 y_1'' + c_2 y_2''] &+ a_1(x)[c_1 y_1' + c_2 y_2'] + a_0(x)[c_1 y_1 + c_2 y_2] \\ &= c_1[a_2(x)y_1'' + a_1(x)y_1' + a_0(x)y_1] + c_2[a_2(x)y_2'' + a_1(x)y_2' + a_0(x)y_2] \\ &= c_1 \cdot 0 + c_2 \cdot 0 = 0. \end{aligned}$$

□

**Corollary 4.4.1.**

1. *A constant multiple $y = c_1 y_1 x)$ of a solution $y_1(x)$ of a homogeneous linear differential equation is also a solution.*
2. *A homogeneous linear differential equation always possesses the trivial solution $y = 0$.*

### 4.4.1 The General Solution to a Linear Second Order Equation

The general solution of (4.1) is given by the linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where $c_1$ and $c_2$ are arbitrary constants and $y_1(x)$ and $y_2(x)$ are linearly independent solutions of (4.1).

## 4.5 The Wronskian

**Definition 4.5.1.** *Let $y_1(x)$ and $y_2(x)$ be any two solutions to the differential equation*

$$y'' + a(x)y' + b(x)y = 0.$$

*The **Wronskian** of $y_1$ and $y_2$ is defined as*

$$W(y_1, y_2) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x).$$

The Wronskian is defined at all points $a$ at which$y_1(x)$ and $y_2(x)$ are differentiable.

**Theorem 4.5.1.** *Two solutions $y_1(x)$ and $y_2(x)$ of (4.1) are linearly independent if and only if $W(y_1, y_2) \neq 0$.*

**Example 4.5.1.** *We see that $y_1(x) = \sin x$ and $y_2(x) = \cos x$ are solutions of $y'' + y = 0$. Using the Wronskian*

$$W(y_1, y_2) = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -\sin^2 x - \cos^2 x = -1 \neq 0.$$

*The functions $y_1(x) = \sin x$ and $y_2(x) = \cos x$ are linearly independent.*

**Example 4.5.2.** *Verify that $y_1(x) = e^{-5x}$ and $y_2(x) = e^{2x}$ are linearly independent solutions of the differential equation*

$$y'' + 3y' - 10y = 0.$$

***Solution:***

$$W(y_1, y_2) = \begin{vmatrix} e^{-5x} & e^{2x} \\ -5e^{-5x} & 2e^{2x} \end{vmatrix} = 2e^{-3x} + 5e^{-3x} = 7e^{-3x} \neq 0.$$

*So the solutions are linearly independent.*

### 4.5.1 Using One Solution to find Another : Reduction of Order

It is easy to write down the general solution of the homogeneous equation

$$y'' + a(x)y' + b(x)y = 0 \tag{4.2}$$

provided we know two linearly independent solutions $y_1$ and $y_2$ of equation (4.2). The general solution is then given by

$$y = c_1y_1 + c_2y_2$$

where $c_1$ and $c_2$ are arbitrary constants.

A procedure exists for finding $y_2$ when $y_1$ is known.

We assume that $y_1$ is a non-zero solution of (4.2) and seek another solution $y_2$ such that $y_1$ and $y_2$ are linearly independent. Since, $y_1$ and $y_2$ are linearly independent, $y_2 \neq ky_1$, for any constant $k$. So $\dfrac{y_2}{y_1} = v(x)$ must be a non constant function of $x$ and $y_2 = vy_1$ must satisfy (4.2). Thus

$$(vy_1)'' + a(vy_1)' + b(vy_1) = 0.$$

But

$$(vy_1)' = vy_1' + v'y_1$$

and

$$(vy_1)'' = (vy_1' + v'y_1)' = vy_1'' + v'y_1' + v''y_1' + v''y_1 = vy_1'' + 2v'y_1' + v''y_1.$$

Then

$$(vy_1'' + 2v'y_1' + v''y_1) + a(vy_1 + v'y_1) + bvy_1 = 0$$

or

$$v(y_1'' + ay_1 + by_1) + v'(2y_1' + ay_1) + v''y_1 = 0.$$

The first term vanishes, since $y_1$ is a solution of (4.2), so we obtain

$$v''y_1 + v'(2y_1' + ay_1) = 0.$$

Dividing by $v'y_1$,

$$\frac{v''}{v'} = -\frac{2y_1'}{y_1} - a.$$

Let $z = v'$, then $z' = v''$ and

$$\frac{z'}{z} = -\frac{2y_1'}{y_1} - a,$$

which is a separable first order equation. Thus we have **reduced the order** of our equation. Integrating with respect to $x$ to obtain

$$\ln z = -2\ln y_1 - \int a(x)dx.$$

Exponentiating, we have

$$z = e^{\ln z} = e^{-2\ln y_1 - \int a(x)dx} = \frac{1}{y_1^2}e^{-\int a(x)dx}.$$

But since $z = v'$,

$$v' = \frac{1}{y_1^2} e^{-\int a(x)dx}.$$

To find $v$, we perform another integration and obtain

$$y_2 = y_1 v = y_1(x) \int \frac{e^{-\int a(x)dx}}{y_1^2(x)} dx.$$

**Example 4.5.3.** *Note that $y_1 = x$ is a solution of*

$$x^2 y'' - xy' + y = 0, \; x > 0. \tag{4.3}$$

*Find the general solution.*

***Solution:*** *Setting $y_2 = y_1 v = xv(x)$, it follows that*

$$y_2' = xv' + v, \quad y_2'' = xv'' + 2v' \quad \textit{and} \quad (4.3)$$

*becomes*

$$x^2(xv'' + 2v') - x(xv' + v) + (xv) = 0 \quad \textit{or} \quad x^3 v'' + x^2 v' = 0.$$

*Setting $z = v'$ and separating variables, we have*

$$\frac{dz}{z} = -\frac{dx}{x} \quad \textit{or} \quad \ln|z| = -\ln x,$$

*and from exponentiation,*

$$v' = z = \frac{1}{x}.$$

*Thus, $v(x) = \ln x$, so $y_2 = y_1 v = x \ln x$ and the general solution of (4.3) is*

$$y = c_1 x + c_2 x \ln x, \quad x > 0.$$

**Example 4.5.4.** *The function $y_1 = x^2$ is a solution of $x^2 y'' - 3xy' + 4y = 0$. Find the general solution on the interval $(0, \infty)$.*

***Solution:*** *Since the equation has the alternative form*

$$y'' - \frac{3}{x} y' + \frac{4}{x^2} y = 0,$$

*we find*

$$\begin{aligned} y_2 &= x^2 \int \frac{e^{3\int dx/x}}{x^4} dx \\ &= x^2 \int \frac{dx}{x} = x^2 \ln x. \end{aligned}$$

*The general solution on $(0, \infty)$ is given by $y = c_1 y_1 + c_2 y_2$, that is,*

$$y = c_1 x^2 + c_2 x^2 \ln x.$$

## 4.6 Homogeneous Equations with Constant Coefficients

### 4.6.1 Case 1: Real Roots

Linear homogeneous equation with constant coefficients is

$$y'' + ay' + by = 0. \tag{4.4}$$

Let a solution of (4.4) be of the form $y(x) = e^{\lambda x}$ for some number $\lambda$ (real or complex), then $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$, so that (4.4) yields

$$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0.$$

Since $e^{\lambda x} \neq 0$, we obtain

$$\lambda^2 + a\lambda + b = 0, \tag{4.5}$$

where $a$ and $b$ are real numbers. Equation (4.5) is called the **characteristic equation** of the differential equation (4.4).

We need to obtain two linearly independent solutions. Equation (4.5) has the two roots

$$\lambda_1 = \frac{-a + \sqrt{a^2 - 4b}}{2} \quad \textit{and} \quad \lambda_2 = \frac{-a - \sqrt{a^2 - 4b}}{2}.$$

Three possibilities arise, $a^2 - 4b > 0$, $a^2 - 4b = 0$ and $a^2 - 4b < 0$.

#### Roots Real and Unequal

If $a^2 - 4b > 0$, then $\lambda_1$ and $\lambda_2$ are distinct real numbers, given by $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$ are distinct solutions.

**Theorem 4.6.1.** *If $a^2 - 4b > 0$, then the roots of the characteristic equation are real and unequal and the general solution to the equation (4.4) is given by*

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

*where $c_1$ and $c_2$ are arbitrary constants and $\lambda_1$ and $\lambda_2$ are the real roots of (4.5).*

**Example 4.6.1.** *Solve the equation*

$$y'' + 3y' - 10y = 0.$$

***Solution:*** *The characteristic equation is $\lambda^2 + 3\lambda - 10 = 0$ and $a^2 - 4b = 49$ and the roots are $\lambda_1 = 2$ and $\lambda_2 = -5$. The general solution is*

$$y(x) = c_1 e^{2x} + c_2 e^{-5x}.$$

*If we specify the initial conditions $y(0) = 1$ and $y'(0) = 3$, for example, differentiating and substituting $x = 0$, we obtain the simultaneous equations*

$$\begin{aligned} c_1 + c_2 &= 1 \\ 2c_1 - 5c_2 &= 3 \end{aligned}$$

*which have the unique solution $c_1 = \dfrac{8}{7}$ and $c_2 = -\dfrac{1}{7}$. The unique solution to the initial value problem is therefore*

$$y(x) = \frac{1}{7}(8e^{2x} - e^{-5x}).$$

**Roots Real and Equal**

Suppose $a^2 - 4b = 0$. We have the double root $\lambda_1 = \lambda_2 = -\dfrac{a}{2}$. Thus $y_1(x) = e^{-\frac{ax}{2}}$ is a solution. To find the second solution $y_2$, we use reduction of order to get

$$\begin{aligned} y_2(x) &= y_1(x) \int \frac{e^{-ax}}{y_1^2} dx \\ &= e^{-\frac{ax}{2}} \int \frac{e^{-ax}}{(e^{-\frac{ax}{2}})^2} dx = e^{-\frac{ax}{2}} \int \frac{e^{-ax}}{e^{-ax}} dx \\ &= e^{-\frac{ax}{2}} \int dx = xe^{-\frac{ax}{2}}. \end{aligned}$$

**Theorem 4.6.2.** *If $a^2 - 4b = 0$, then the roots of the characteristic equation are equal and the general solution is given by*

$$y(x) = c_1 e^{-\frac{ax}{2}} + c_2 x e^{-\frac{ax}{2}}$$

*where $c_1$ and $c_2$ are arbitrary constants.*

**Example 4.6.2.** *Solve the equation*

$$y'' - 6y' + 9y = 0.$$

***Solution:*** *The characteristic equation is $\lambda^2 - 6\lambda + 9 = 0$ and $a^2 - 4b = 0$, yielding the unique double root, $\lambda_1 = -\dfrac{a}{2} = 3$. The general solution is*

$$y(x) = c_1 e^{3x} + c_2 x e^{3x}.$$

*If we use the initial conditions $y(0) = 1$ and $y'(0) = 7$, we obtain the simultaneous equations*

$$c_1 = 1, \quad 3c_1 + c_2 = 7 \Rightarrow c_2 = 4.$$

*Then the solution for this initial value problem is*

$$y(x) = e^{3x} + 4xe^{3x} = e^{3x}(1 + 4x).$$

**Complex Conjugate Roots**

We have from Eulers' formula
$$e^{i\beta x} = \cos\beta x + i\sin\beta x.$$
Suppose $a^2 - 4b < 0$. The roots of the characteristic equation are
$$\lambda_1 = \alpha + i\beta, \quad \lambda_2 = \alpha - i\beta.$$

**Theorem 4.6.3.** *If $a^2 - 4b < 0$, then the characteristic equation has complex conjugate roots and the general solution is given by*
$$y(x) = c_1 e^{\alpha x}\cos\beta x + c_2 e^{\alpha x}\sin\beta x.$$

**Example 4.6.3.** *Solve $y'' + y = 0$.*

***Solution:*** *The characteristic equation is $\lambda^2 + 1 = 0$, with roots $\pm i$. We have $\alpha = 0$ and $\beta = 1$, so the general solution is*
$$y(x) = c_1\cos x + c_2\sin x.$$

## 4.7 Non-Homogeneous Equations

**Definition 4.7.1.** *A solution $y_p$ is a **particular solution** or **particular integral** of a differential equation if $y_p$ has no arbitrary parameters or constants.*

Let $y'' + ay' + by = f(x)$ be a non-homogeneous linear differential equation. Then $y'' + ay' + by = 0$ is the associated homogeneous linear differential equation.

**Theorem 4.7.1.** *Let $y_p$ be any particular solution of a non-homogeneous differential equation and let $\{y_1, y_2, \cdots, y_n\}$ be a fundamental set of solutions of the associated homogeneous differential equation. Then,the general solution of the non-homogeneous equation is*
$$y(x) = c_1y_1 + c_2y_2 + \cdots + c_ny_n + y_p$$
*where $c_i, i = 1, 2, \ldots, n$ are arbitrary constants.*

The part $c_1y_1 + c_2y_2 + \cdots + c_ny_n$ is called the **complementary function**. So, in general, the solution of a non-homogeneous linear differential equation is
$$y(x) = \textit{complementary solution} + \textit{any particular solution}.$$

The **method of undetermined coefficients** is used to find particular solutions to a given non-homogeneous linear second order differential equation, either having

(a) a polynomial including constants.

(b) a constant multiplied by the sine and cosine function and their linear combination.

(c) exponential functions.

(d) linear combination of functions in (a),(b) and (c).

| $f(x)$ | Appropriate form of $y_p$ |
|---|---|
| 1 or any constant | $A$ |
| $ax+b$ | $Ax+B$ |
| $ax^2+bx+c$ | $Ax^2+bx+C$ |
| $\sin kx$ | $A\sin kx + B\cos kx$ |
| $\cos kx$ | $A\sin kx + B\cos kx$ |
| $ae^{kx}$ | $Ae^{kx}$ |
| $(ax+b)e^{kx}$ | $(Ax+B)e^{kx}$ |
| $x^2e^{kx}$ | $(Ax^2+Bx+C)e^{kx}$ |
| $e^{kx}\sin mx$ | $Ae^{kx}\sin mx + Be^{kx}\cos mx$ |

**Example 4.7.1.** *Solve the following equation*

$$y'' - y = x^2$$

***Solution:*** *Here $f(x) = x^2$ is a polynomial of degree two, so $y_p$ is also a polynomial of degree two. We try a general polynomial of degree two, say $y_p(x) = a + bx + cx^2$, then $y_p'(x) = b + 2cx$ and $y_p''(x) = 2c$. Substituting, we obtain*

$$2c - (a + bx + cx^2) = x^2.$$

*Equating coefficients*

$$\begin{aligned} coefficients\,of\,constant\,term : 2c - a &= 0 \\ of\,x : -b &= 0 \\ of\,x^2 : -c &= 1, \end{aligned}$$

*which yields $a = -2$, $b = 0$ and $c = -1$ and the particular solution is*

$$y_p(x) = -2 - x^2.$$

*General solution of the homogeneous equation $y'' - y = 0$ is given by $y = c_1e^x + c_2e^{-x}$ and the general solution of the non-homogeneous equation is*

$$y = c_1e^x + c_2e^{-x} - 2 - x^2.$$

**Example 4.7.2.** *Solve*

$$y'' - 3y' + 2y = e^x \sin x.$$

***Solution:*** *The particular solution is of the form* $y_p(x) = ae^x \sin x + be^x \cos x$*, then* $y_p' = (a-b)e^x \sin x + (a+b)e^x \cos x$ *and* $y_p'' = 2ae^x \cos x - 2be^x \sin x$*. Substituting, we have*

$$e^x(2a\cos x - 2b\sin x) - 3e^x[(a-b)\sin x + (a+b)\cos x] + 2e^x(a\sin x + b\cos x) = e^x \sin x.$$

*Dividing both sides by* $e^x$ *and equating the coefficients of* $\sin x$ *and* $\cos x$*, we have*

$$\begin{aligned} 2a - 3(a+b) + 2b &= 0 \\ -2b - 3(a-b) + 2a &= 1 \end{aligned}$$

*which yields* $a = -\dfrac{1}{2}$ *and* $b = \dfrac{1}{2}$*, so that*

$$y_p = \frac{e^x}{2}(\cos x - \sin x).$$

*Finally, the general solution is*

$$y = c_1 e^{2x} + c_2 e^x + \frac{e^x}{2}(\cos x - \sin x).$$

**Example 4.7.3.** *Solve*

$$y'' + y = xe^{2x}.$$

***Solution:*** *The particular solution is of the form* $y_p(x) = e^{2x}(a+bx)$*, then* $y_p'(x) = e^{2x}(2a+b+2bx)$ *and* $y_p''(x) = e^{2x}(4a+4b+4bx)$ *and substituting yields*

$$e^{2x}(4a+4b+4bx) + e^{2x}(a+bx) = xe^{2x}.$$

*Dividing both sides by* $e^{2x}$ *and equating like powers, we have*

$$5a + 4b = 0,\ 5b = 1.$$

*Thus* $a = -\dfrac{4}{25}$*,* $b = \dfrac{1}{5}$ *and a particular solution is*

$$y_p(x) = \frac{e^{2x}}{25}(5x - 4).$$

*Therefore the general solution is*

$$y(x) = c_1 \sin x + c_2 \cos x + \frac{e^{2x}}{25}(5x-4).$$

**Example 4.7.4.** *Find the solution to the equation*

$$y'' - y = 2e^x.$$

***Solution:*** *The general solution of* $y'' - y = 0$ *is*

$$y(x) = c_1 e^x + c_2 e^{-x}.$$

*Then* $y_p = Axe^x$*, and* $y_p' = Ae^x(x+1)$*,* $y_p'' = Ae^x(x+2)$ *and substituting, we have*

$$y_p'' - y_p = Ae^x(x+2) - Axe^x = 2Ae^x = 2e^x.$$

*Here* $A = 1$ *and* $y_p = xe^x$*. Thus, the general solution is*

$$y(x) = c_1 e^x + c_2 e^{-x} + xe^x.$$

Given the homogeneous equation

$$y'' + ay' + by = 0. \tag{4.6}$$

If any term of the guessed solution $y_p(x)$ is a solution of the homogeneous equation (4.6), multiply $y_p(x)$ by $x$ repeatedly until no term of the product $x^k y_p(x)$ is a solution of (4.6). Then use the product $x^k y_p(x)$ to solve the equation

$$y'' + ay' + by = f(x).$$

**Example 4.7.5.** *Find the solution to*

$$y'' + y = \cos x$$

*that satisfies $y(0) = 2$ and $y'(0) = -3$.*

***Solution:*** *The general solution to $y'' + y = 0$ is $y = c_1 \cos x + c_2 \sin x$. Since $y = \cos x$ is a solution. Ordinarily we would guess a solution of the form*

$$y_p = A \cos x + B \sin x.$$

*Instead we multiply by $x$ and try a solution of the form*

$$y_p = Ax \cos x + Bx \sin x.$$

*Note no terms of $y_p$ is a solution of $y'' + y = 0$. Then $y_p' = A \cos x - Ax \sin x + B \sin x + Bx \cos x$ and*

$$\cos x = y_p'' + y_p = (-2A \sin x - Ax \cos x + 2B \cos x - Bx \sin x) + (Ax \cos x + Bx \sin x) = -2A \sin x + 2B \cos x.$$

*Therefore $-2A = 0$, $2B = 1 \Rightarrow b = \dfrac{1}{2}$ and $y_p = \dfrac{1}{2} x \sin x$. The general solution is*

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{2} x \sin x.$$

*We have*

$$y' = -c_1 \sin x + c_2 \cos x + \frac{1}{2} x \cos x + \frac{1}{2} \sin x.$$

*Then $y(0) = c_1 = 2$ and $y'(0) = c_2 = -3$, which yields the unique solution*

$$y(x) = 2 \cos x - 3 \sin x + \frac{1}{2} x \sin x.$$

**Example 4.7.6.** *Find the general solution of*

$$y'' - 4y' + 4y = e^{2x}.$$

***Solution:*** *The homogeneous equation $y'' - 4y' + 4y = 0$ has the independent solutions $e^{2x}$ and $xe^{2x}$. Multiplying $e^{2x}$ by $x$ twice, we look for a particular solution of the form $y_p = ax^2 e^{2x}$. Then $y_p' = ae^{2x}(2x^2 + 2x)$ and $y_p'' = ae^{2x}(4x^2 + 8x + 2)$, so*

$$y_p'' - 4y_p' + 4y_p = ae^{2x}(4x^2 + 8x + 2 - 8x^2 - 8x + 4x^2) = 2ae^{2x} = e^{2x}$$

*or $2a = 1$ and $a = \dfrac{1}{2}$. Thus $y_p = \dfrac{1}{2} x^2 e^{2x}$ and the general solution is*

$$y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{2} x^2 e^{2x} = e^{2x} \left( c_1 + c_2 x + \frac{1}{2} x^2 \right).$$

## 4.8 Higher Order Equations

**Example 4.8.1.** *Solve* $y''' + 3y'' - 4y = 0$.

***Solution:*** *From the characteristic equation* $\lambda^3 + 3\lambda^2 - 4 = 0$*, that one root is* $q\lambda_1 = 1$*. Dividing* $\lambda^3 + 3\lambda^2 - 4$ *by* $\lambda - 1$*, we find*

$$\lambda^3 + 3\lambda^2 - 4 = (\lambda - 1)(\lambda^2 + 4\lambda + 4) = (\lambda - 1)(\lambda + 2)^2,$$

*and so the other roots are* $\lambda_2 = \lambda_3 = -2$*. Thus the general solution is*

$$y = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}.$$

## 4.9 The Method of Variation of Parameters

This method of finding a particular solution is due to J.L. Lagrange (1736-1813). It consists in using the general solution of the homogeneous equation. The variation of parameter method can be applied to more general equations than the undetermined coefficients method.

**Theorem 4.9.1.** *Let* $y_1$ *and* $y_2$ *be linearly independent solutions to the homogeneous equation*

$$y'' + p(x)y' + q(x)y = 0$$

*and let* $W = y_1 y_2' - y_1' y_2$*. If* $u_1$ *and* $u_2$ *are defined by*

$$u_1(x) = \int \frac{-y_2(x)f(x)}{W}dx, \quad u_2(x) = \int \frac{y_1(x)f(x)}{W}dx,$$

*then the function* $y_p = u_1 y_1 + u_2 y_2$ *is a particular solution to the non-homogeneous equation*

$$y'' + p(x)y' + q(x)y = f(x).$$

**Example 4.9.1.** *Find the general solution of* $y'' - 5y' + 6y = 2e^x$.

***Solution:*** *The characteristic equation is*

$$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda_1 = 3 \quad \text{and} \quad \lambda_2 = 2.$$

*Hence* $y_1 = e^{3x}$ *and* $y_2 = e^{2x}$*. Then* $W = e^{3x}(2e^{2x}) - (3e^{3x})e^{2x} = -e^{5x}$*. Compute the functions* $u_1$ *and* $u_2$*. By definition*

$$u_1' = \frac{-y_2 f}{W} \quad \text{and} \quad u_2' = \frac{y_1 f}{W}.$$

*So*

$$\begin{aligned} u_1' &= -e^{2x}2e^x(-e^{-5x}) = 2e^{-2x} \Rightarrow u_1 = -e^{-2x} \\ u_2' &= e^{3x}(2e^x)(-e^{-5x}) = -2e^{-x} \Rightarrow u_2 = 2e^{-x}. \end{aligned}$$

*The particular solution is*

$$y_p = (-e^{-2x})(e^{3x}) + (2e^{-x})(e^{2x}) = e^x.$$

*The general solution is*

$$y = c_1e^{3x} + c_2e^{2x} + e^x.$$

**Example 4.9.2.** *Solve* $y'' - 4y' + 4y = (x+1)e^{2x}$.

***Solution:*** *From the characteristic equation* $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0$, *we have* $y_h = c_1e^{2x} + c_2xe^{2x}$. *Then* $y_1 = e^{2x}$ *and* $y_2 = xe^{2x}$. *Then* $W = e^{2x}(2xe^{2x} + e^{2x}) - 2e^{2x}(xe^{2x}) = e^{4x}$. *We identify* $f(x) = (x+1)e^{2x}$. *Then*

$$\begin{aligned} u_1' &= \frac{-(x+1)xe^{4x}}{e^{4x}} = -x^2 - x, \\ u_2' &= \frac{(x+1)e^{4x}}{e^{4x}} = x + 1. \end{aligned}$$

*It follows that* $u_1 = -\frac{1}{3}x^3 - \frac{1}{2}x^2$ *and* $u_2 = \frac{1}{2}x^2 + x$. *Hence*

$$y_p = \left(-\frac{1}{3}x^3 - \frac{1}{2}x^2\right)e^{2x} + \left(\frac{1}{2}x^2 + x\right)xe^{2x} = \frac{1}{6}x^3e^{2x} + \frac{1}{2}x^2e^{2x}$$

*and*

$$y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{6}x^3e^{2x} + \frac{1}{2}x^2e^{2x}.$$

## 4.10 Cauchy-Euler Equation

A linear differential equation of the form

$$a_nx^n\frac{d^ny}{dx^n} + a_{n-1}x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1x\frac{dy}{dx} + a_0y = g(x),$$

where the coefficients $a_n, a_{n-1}, \ldots, a_0$ are constants, is known as a **Cauchy-Euler equation**. The observable characteristic of this type of equation is that the degree $k = n, n-1, \ldots, 1, 0$ of the monomial coefficients $x^k$ matches the order $k$ of differentiation $\dfrac{d^ky}{dx^k}$.

To start, we confine our attention to solving the homogeneous second order equation

$$ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = 0.$$

The solution of higher order equations follow analogously. Also we can solve the non-homogeneous equation

$$ax^2y'' + bxy' + cy = g(x)$$

by variation of parameters, once we have determined the complementary function $y_c(x)$.

### 4.10.1 Method of Solution

We try a solution of the form $y = x^m$, where $m$ is to be determined. The first and second derivatives are, respectively,

$$\frac{dy}{dx} = mx^{m-1}, \quad \text{and} \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}.$$

Consequently

$$\begin{aligned} ax^2\frac{d^2y}{dx^2} + dx\frac{dy}{dx} + cy &= ax^2 \cdot m(m-1)x^{m-2} + bx \cdot mx^{m-1} + cx^m \\ &= am(m-1)x^m + bmx^m + cx^m = x^m(am(m-1) + bm + c). \end{aligned}$$

Thus $y = x^m$ is a solution of the differential equation whenever $m$ is a solution of the **auxiliary equation**

$$am(m-1) + bm + c = 0 \quad \text{or} \quad am^2 + (b-a)m + c = 0.$$

There are three different cases to be considered, depending on whether the roots of this quadratic equation are real and distinct, real and equal, or complex.

#### CASE 1 Distinct Real Roots

Let $m_1$ and $m_2$ denote the real roots and $m_1 \neq m_2$. Then $y_1 = x^{m_1}$ and $y_2 = x^{m_2}$ form a fundamental set of solutions. Hence the general solution is

$$y = c_2x^{m_1} + c_2x^{m_2}.$$

**Example 4.10.1.** *Solve* $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0.$

***Solution:*** *The substitution* $y = x^m$ *yields*

$$\begin{aligned} x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y &= x^2 \cdot m(m-1)x^{m-2} - 2x \cdot mx^{m-1} - 4x^m \\ &= x^m(m(m-1) - 2m - 4) = x^m(m^2 - 3m - 4) = 0 \end{aligned}$$

*if* $m^2 - 3m - 4 = 0$. *Now* $(m+1)(m-4) = 0$ *implies* $m_1 = -1, m_2 = 4$ *so that*

$$y = c_1x^{-1} + cx^4.$$

#### CASE 2 Repeated Roots

If the roots are repeated, that is, $m_1 = m_2$, then the general solution is

$$y = c_1x^{m_1} + c_2x^{m_1}\ln x.$$

**Example 4.10.2.** *Solve* $4x^2\dfrac{d^2y}{dx^2}+8x\dfrac{dy}{dx}+y=0$.

***Solution:*** *The substitution* $y=x^m$ *yields*

$$4x^2\frac{d^2y}{dx^2}+8x\frac{dy}{dx}+y=x^m(4m(m-1)+8m+1)=x^m(4m^2+4m+1)=0$$

*when* $4m^2+4m+1=0$ *or* $(2m+1)^2=0$. *Since* $m_1=-\frac{1}{2}$, *the general solution is*

$$y=c_1x^{-\frac{1}{2}}+c_2x^{-\frac{1}{2}}\ln x.$$

For higher order equations, if $m_1$ is a root of multiplicity $k$, then it can be shown that

$$x^{m_1}, x^{m_1}\ln x, x^{m_1}(\ln x)^2, \ldots, x^{m_1}(\ln x)^{k-1}$$

are $k$ linearly independent solutions. Correspondingly, the general solution of the differential equation must then contain a linear combination of these $k$ solutions.

### CASE 3 Conjugate Complex Roots

If the roots are the conjugate pair $m_1=\alpha+i\beta$, $m_2=\alpha-i\beta$, where $\alpha$ and $\beta>0$ are real, then the general solution is

$$y=x^\alpha[c_1\cos(\beta\ln x)+c_2\sin(\beta\ln x)].$$

**Example 4.10.3.** *Solve the initial value problem*

$$x^2\frac{d^2y}{dx^2}+3x\frac{dy}{dx}+3y=0,\ y(1)=1,\ y'(1)=-5.$$

***Solution:*** *We have*

$$x^2\frac{d^2y}{dx^2}+3x\frac{dy}{dx}+3y=x^m(m(m-1)+3m+3)=x^m(m^2+2m+3)=0$$

*when* $m^2+2m+3=0$. *From the quadratic formula we find* $m_1=-1+\sqrt{2}i$ *and* $m_2=-1-\sqrt{2}i$. *If we make the identifications* $\alpha=-1$ *and* $\beta=\sqrt{2}$, *we see that the general solution of the differential equation is*

$$y=x^{-1}[c_1\cos(\sqrt{2}\ln x)+c_2\sin(\sqrt{2}\ln x)].$$

*By applying the conditions* $y(1)=1, y'(1)=-5$ *to the foregoing solution, we find, in turn,* $c_1=1$ *and* $c_2=-2\sqrt{2}$. *Thus the solution to the initial value problem is*

$$y=x^{-1}[\cos(\sqrt{2}\ln x)-2\sqrt{2}\sin(\sqrt{2}\ln x)].$$

The next example illustrates the solution of a third order Cauchy-Euler equation.

**Example 4.10.4.** *Solve* $x^3\dfrac{d^3y}{dx^3}+5x^2\dfrac{d^2y}{dx^2}+7x\dfrac{dy}{dx}+8y=0.$

***Solution:*** *The first three derivatives of $y=x^m$ are*

$$\frac{dy}{dx}=mx^{m-1},\quad \frac{d^2y}{dx^2}=m(m-1)x^{m-2},\quad \frac{d^3y}{dx^3}=m(m-1)(m-2)x^{m-3},$$

*so that the given differential equation becomes*

$$\begin{aligned} x^3\frac{d^3y}{dx^3}+5x^2\frac{d^2y}{dx^2}+7x\frac{dy}{dx}+8y &= x^3m(m-1)(m-2)x^{m-3}+5x^2m(m-1)x^{m-2}+7xmx^{m-1}+8x^m \\ &= x^m(m(m-1)(m-2)+5m(m-1)+7m+8) \\ &= x^m(m^3+2m^2+4m+8)=x^m(m+2)(m^2+4)=0. \end{aligned}$$

*In this case we see that $y=x^m$ will be a solution of the differential equation for $m_1=-2, m_2=2i$ and $m_3=-2i$. Hence the general solution is*

$$y=c_1x^{-2}+c_2\cos(2\ln x)+c_3\sin(2\ln x).$$

Because the method of undetermined coefficients is applicable only to differential equations with constant coefficients its cannot be applied directly to a non-homogeneous Cauchy-Euler equation.

### 4.10.2 Using Variation of Parameters

**Example 4.10.5.** *Solve the non-homogeneous equation*

$$x^2y''-3xy'+3y=2x^4e^x.$$

***Solution:*** *The substitution $y=x^m$ leads to the auxiliary equation*

$$m(m-1)-3m+3=0 \quad \textit{or} \quad (m-1)(m-3)=0.$$

*Thus*

$$y_c=c_1x+c_2x^3.$$

*Therefore, from*

$$y''-\frac{3}{x}y'+\frac{3}{x^2}y=2x^2e^x$$

*we make the identification $f(x)=2x^2e^x$. Now with $y_1=x, y_2=x^3$, and*

$$W=\begin{vmatrix} x & x^3 \\ 13x^2 & \end{vmatrix}=2x^3,\quad W_1=\begin{vmatrix} 0 & x^3 \\ 2x^2e^x & 3x^2 \end{vmatrix}=-2x^5e^x,\quad W_2=\begin{vmatrix} x & 0 \\ 1 & 2x^2e^x \end{vmatrix}=2x^3e^x$$

*we find $u_1'=-\dfrac{2x^5e^x}{2x^3}=-xe^x$ and $u_2'=\dfrac{2x^3e^x}{2x^3}=e^x$. The integral of the latter function is immediate, but in case of $u_1'$, we integrate by parts twice. The results are $u_1=-x^2e^x+2xe^x-2e^x$ and $u_2=e^x$. Hence*

$$y_p=u_1y_1+u_2y_2=(-x^2e^x+2xe^x-2e^x)x+x^3e^x=2x^2e^x-2xe^x.$$

*Finally we have*

$$y=y_c+y_p=c_1x+c_2x^3+2x^2e^x-2xe^x.$$

## 4.11 Problems

1. Solve the given differential equation by undetermined coefficients.
(i) $y'' + 3y' + 2y = 6$ (ii) $y'' - 8y' + 20y = 100x^2 - 26xe^x$ (iii) $y'' + 4y = 3\sin 2x$
(iv) $y'' - 2y' + 5y = e^x \cos 2x$ (v) $y'' + 2y' - 24y = 16 - (x+2)e^{4x}$
(vi) $y''' - y'' - 4y' + 4y = 5 - e^x + e^{2x}$ (vii) $y^{(4)} - y'' = 4x + 2xe^{-x}$
(viii) $y''' - 2y'' + 3y' - y = x - 4e^x$.

2. Solve the given differential equation subject to the indicated initial conditions.
(i) $2y'' + 3y' - 2y = 14x^2 - 4x - 11$, $y(0) = 0$, $y'(0) = 0$
(ii) $5y'' + y' = -6x$, $y(0) = 0$, $y'(0) = -10$
(iii) $y''' + 8y = 2x - 5 + 8e^{-2x}$, $y(0) = -5$, $y'(0) = 3$, $y''(0) = -4$.

3. Solve each differential equation by variation of parameters.
(i) $y'' + y = \sec x$ (ii) $y'' - 3y' + 2y = \dfrac{e^{3x}}{1+e^x}$ (iii) $y''' + y' = \tan x$
(iv) $y'' + 2y' + y = e^{-x}\ln x$ (v) $4y'' - 4y' + y = e^{\frac{x}{2}}\sqrt{1-x^2}$.

4. Solve the given differential equation.
(i) $x^2y'' - 2y = 0$ (ii) $3x^3y'' + 6xy' + y = 0$ (iii) $x^3y''' + xy' - y = 0$ (iv) $xy'' - y' = 0$.

5. Solve by variation of parameters.
(i) $xy'' + y' = x$ (ii) $x^2y'' - 2xy' + 2y = x^4e^x$ (iii) $x^2y'' - 2xy' + 2y = x^3\ln x$.

6. Any Cauchy-Euler differential equation can be reduced to an equation with constant coefficients by means of the substitution $x = e^t$. Solve the given differential equation by means of this substitution.
(i) $x^2y'' + 10xy' + 8y = x^2$ (ii) $x^2y'' - 4xy' + 6y = \ln x^2$ (iii) $x^2y'' + 9xy' - 20y = \dfrac{5}{x^3}$.

7. Solve the initial-value problem
$$\frac{d^2x}{dt^2} + \omega^2 x = F_0 \sin\gamma t, \quad x(0) = x'(0) = 0$$
where $F_0$ is a constant and $\gamma \neq \omega$. Find the solution when $\gamma = \omega$ by taking the limit of the solution obtained as $\gamma \to \omega$.

8. The equation representing the vibration system consisting of mass $m$, a spring with spring constant $k$, a damping constant $\beta$ and an external force $F(t)$, is given by
$$m\frac{d^2x}{dt^2} + \beta\frac{dx}{dt} + kx = F(t)$$
where $x$ is the displacement measured from equilibrium. Given that $m = 0.2$, $\beta = 1.2$, $k = 2$ and $F(t) = 5\cos 4t$, solve the above subject to $x(0) = \frac{1}{2}$ and $x'(0) = 0$. Show that the solution of the homogeneous part $x_h$ has the property $x_h(t) \to 0$ as $t \to \infty$.

9. The temperature $u(r)$ in the circular ring is determined from the boundary value problem
$$r\frac{d^2u}{dr^2} + \frac{du}{dr} = 0, \quad u(a) = u_0, \quad u(b) = u_1,$$

where $u_0$ and $u_1$ are constants. Show that

$$u(r) = \frac{u_0 \ln(r/b) - u_1 \ln(r/a)}{\ln(a/b)}.$$

10. Consider two concentric spheres of radius $r = a$ and $r = b$, $a < b$. The temperature $u(r)$ in the region between the spheres is determined from the boundary value problem

$$r\frac{d^2u}{dr^2} + 2\frac{du}{dr} = 0, \quad u(a) = u_0, \quad u(b) = u_1,$$

where $u_0$ and $u_1$ are constants. Solve for $u(r)$.

# Chapter 5

# Existence and Uniqueness of Solutions

*If you tell the truth, you don't have to remember anything.*

—Mark Twain

## 5.1 Introduction

So far we have engaged ourselves in solving differential equations, tacitly assuming that there always exists a solution. However, the theory of existence and uniqueness of solutions of the initial value problems is quite complex. We begin to develop this theory for the initial value problem

$$y' = f(x, y),\ y(x_0) = y_0, \tag{5.1}$$

where $f(x, y)$ will be assumed to be continuous in a domain $D$ containing the point $(x_0, y_0)$. By a solution of (5.1) in an interval $J$ containing $x_0$, we mean a function $y(x)$ satisfying

(i) $y(x_0) = y_0$

(ii) $y'(x)$ exists for all $x \in J$

(iii) for all $x \in J$ the points $(x, y(x)) \in D$

(iv) $y'(x) = f(x, y(x))$ for all $x \in J$.

**Theorem 5.1.1** (Existence)**.** *Suppose that $f(x, y)$ is a continuous function defined in some region*

$$D = \{(x, y) : x_0 - \delta < x < x_0 + \delta, \quad y_0 - \epsilon < y < y_0 + \epsilon\}$$

*containing the point $(x_0, y_0)$. Then, there exists a number $\delta_1$ (possibly smaller than $\delta$) so that a solution $y = f(x)$ to (5.1) is defined for $x_0 - \delta_1 < x < x_0 + \delta_1$.*

**Theorem 5.1.2** (Uniqueness)**.** *Suppose that both $f(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ are continuous functions defined on a region $D$. Then there exists a number $\delta_2$ (possibly smaller than $\delta_1$) so that the solution $y = f(x)$ to (5.1), whose existence was guaranteed by the above Theorem, is the unique solution to (5.1) for $x_0 - \delta_2 < x < x_0 + \delta_2$.*

**Example 5.1.1.** *Consider the ordinary differential equation*

$$y' = x - y + 1, \quad y(1) = 2.$$

*In this case, both the function $f(x, y) = x - y + 1$ and its partial derivative $\frac{\partial f}{\partial y}(x, y) = -1$ are defined and continuous at all points $(x, y)$. The Theorem guarantees that a solution to the ordinary differential equation exists in some **open interval** centered at 1, and that this solution is unique in some (possibly smaller) interval centered at 1. In fact, an explicit solution to this equation is*

$$y(x) = x + e^{1-x}.$$

*(Check this for yourself. This solution exists (and is the unique solution to the equation) for all real numbers $x$.*

**Example 5.1.2.** *Consider the ordinary differential equation*

$$y' = 1 + y^2, \quad y(0) = 0.$$

*Again, both $f(x, y) = 1 + y^2$ and $\frac{\partial f}{\partial y}(x, y) = 2y$ are defined and continuous at all points $(x_0, y_0)$, so by the Theorem, we can conclude that a solution exists in some open interval centered at 0, and is unique in some (possibly smaller) interval centered at 0.*

*By separating variables and integrating, we derive a solution to this equation of the form*

$$y(x) = \tan x.$$

*As an abstract function of $x$, this defined for all $x \neq \dots, \frac{-3\pi}{2}, \frac{-\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots$. However, in order for this function to be considered as a solution to this ordinary differential equation we must restrict the domain. (**Remember that a solution to a differential equation must be a continuous function!**) Specifically, the function*

$$y = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

*is a solution to the above ordinary differential equation.*

**Example 5.1.3.** *Consider the ordinary differential equation*

$$y' = \frac{2y}{x}, \quad y(x_0) = y_0.$$

*In this example, $f(x, y) = \frac{2y}{x}$ and $\frac{\partial f}{\partial y}(x, y) = \frac{2}{x}$. Both of these functions are defined for all $x \neq 0$, so by the Theorem, tells us that for $x_0 \neq 0$ there exists a unique solution defined in an open interval around $x_0$. By separating variables and integrating, we derive solutions to this equation of the form*

$$y(x) = Cx^2$$

*for any constant $C$. Note that all of these solutions pass through the point $(0,0)$ and that none of them pass through any point $(0, y_0)$ with $y_0 \neq 0$. So the initial value problem*

$$y' = \frac{2y}{x}, \quad y(0) = 0$$

*has infinitely many solutions, but the initial value problem*

$$y' = \frac{2y}{x}, \quad y(0) = y_0, \quad y_0 \neq 0$$

*has no solutions.*

*For each $(x_0, y_0)$ with $x_0 \neq 0$, there is a unique parabola $y = Cx^2$ whose graph passes through $(x_0, y_0)$. So the initial value problem*

$$y' = \frac{2y}{x}, \quad y(x_0) = y_0, \quad x_0 \neq 0$$

*has a unique solution.*

*The initial value problem $y(x_0) = y_0$ has*

- *a unique solution in an open interval containing $x_0$ if $x_0 \neq 0$.*
- *no solution if $x_0 = 0$ and $y_0 \neq 0$.*
- *infinitely many solutions if $(x_0, y_0) = (0, 0)$.*

# Chapter 6

# Series Solutions

*A lie can travel half way around the world while the truth is putting on its shoes.*

—Mark Twain

## 6.1 Review of Power Series

We have seen that solving a homogeneous linear differential equation with constant coefficients was essentially a problem in algebra. By finding the roots of the auxiliary equation, we could write a general solution of the differential equation as a linear combination of the elementary functions $x^k, x^k e^{\alpha x}, x^k e^{\alpha x}\cos\beta x$, and $x^k e^{\alpha x}\sin\beta x$, where $k$ is a non-negative integer. But most linear higher-order differential equations with variable coefficients cannot be solved in terms of elementary functions. A usual course of action for equations of this sort is to assume a solution in the form of infinite series and proceed in a manner similar to the method of undetermined coefficients. In this chapter we consider linear second-order differential equations with variable coefficients that possess solutions in the form of power series.

- **Definition of a Power Series** A **power series** in $x - a$ is an infinite series of the form $\displaystyle\sum_{n=0}^{\infty} c_n(x-a)^n$. A series such as this is also said to be a **power series centered at** $a$.

  **Example 6.1.1.** $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}x^n$ *is a power series in $x$ and centered at zero.*

- **Convergence** If the series equals a finite real constant for the given $x$, then the series is said to **converge** at $x$. If the series does not converge at $x$, it is said to **diverge** at $x$.

- **Interval of Convergence** Every power series has an interval of convergence. The interval of convergence is the set of all numbers for which the series converges.

- **Radius of Convergence** Every interval of convergence has a **radius of convergence** $R$.
- **Absolute Convergence** Within its interval of convergence a power series converges absolutely. In other words, for $x$ in the interval of convergence the series of absolute values $\sum_{n=0}^{\infty} |c_n||(x-a)^n|$ converges.
- **A Power Series Represents a Function** A power series represents a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots$$

whose domain is the interval of convergence of the series.

## 6.2 Power Series Solution of a Differential Equation

**Example 6.2.1.** *Find a solution of*

$$\frac{dy}{dx} - 2xy = 0$$

*in the form of a power series in $x$.*

***Solution:*** *If we assume that a solution of the given equation exists in the form*

$$y = \sum_{n=0}^{\infty} c_n x^n, \tag{6.1}$$

*we pose the question, "**Can we determine coefficients** $c_n$ **for which the power series converges to a function satisfying** $\frac{dy}{dx} - 2xy = 0$? Formal term by term differentiation of* (6.1) *gives*

$$\frac{dy}{dx} = \sum_{n=0}^{\infty} nc_n x^{n-1} = \sum_{n=1}^{\infty} nc_n x^{n-1}.$$

*Note that since the first term in the first series (corresponding to $n = 0$) is zero, we begin summation with $n = 1$. Using the last result and assumption* (6.1), *we find*

$$\frac{dy}{dx} - 2xy = \sum_{n=1}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} 2c_n x^{n+1}. \tag{6.2}$$

*We would like to add the two series in* (6.2). *To this end we write*

$$\frac{dy}{dx} - 2xy = 1 \cdot c_1 x^0 + \sum_{n=2}^{\infty} nc_n x^{n-1} - \sum_{n=0}^{\infty} 2c_n x^{n+1} \tag{6.3}$$

*and the by letting $k = n - 1$ in the first series and $k = n + 1$ in the second, the right side of* (6.3) *becomes*

$$c_1 + \sum_{k=1}^{\infty} (k+1)c_{k+1} x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k.$$

*After we add the series term-wise, it follows that*

$$\frac{dy}{dx} - 2xy = c_1 + \sum_{k=1}^{\infty} \left[(k+1)c_{k+1} - 2c_{k-1}\right] x^k = 0. \tag{6.4}$$

*Hence in order to have* (6.4) *identically zero, it is necessary that the coefficients of like powers of* $x$ *be zero, that is,*

$$c_1 = 0 \quad \textit{and} \quad (k+1)c_{k+1} - 2c_{k-1} = 0,\ k = 1, 2, 3, \ldots \tag{6.5}$$

*Equation* (6.5) *provides a* ***recurrence relation*** *that determines the* $c_k$. *Since* $k + 1 \neq 0$*for all indicated values of* $k$*, we can write* (6.5) *as*

$$c_{k+1} = \frac{2c_{k-1}}{k+1}. \tag{6.6}$$

*Iteration of this formula, then gives*

$$\begin{aligned}
k = 1, &\quad c_2 = \frac{2}{2}c_0 = c_0 \\
k = 2, &\quad c_3 = \frac{2}{3}c_1 = 0 \\
k = 3, &\quad c_4 = \frac{2}{4}c_2 = \frac{1}{2}c_0 = \frac{1}{2!}c_0 \\
k = 4, &\quad c_5 = \frac{2}{5}c_3 = 0 \\
k = 5, &\quad c_6 = \frac{2}{6}c_4 = \frac{1}{3 \cdot 2!}c_0 = \frac{1}{3!}c_0 \\
k = 6, &\quad c_7 = \frac{2}{7}c_5 = 0 \\
k = 7, &\quad c_8 = \frac{2}{8}c_6 = \frac{1}{4 \cdot 3!}c_0 = \frac{1}{4!}c_0
\end{aligned}$$

*and so on. Thus from the original assumption* (6.1)*, we find*

$$\begin{aligned}
y = \sum_{n=0}^{\infty} c_0 x^n &= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c^4 x^4 + c_5 x^5 + \cdots \\
&= c_0 + 0 + c_0 x^2 + 0 + \frac{1}{2!}c_0 x^4 + 0 + \frac{1}{3!}c_0 x^6 + 0 + \cdots \\
&= c_0 \left[1 + x^2 + \frac{1}{2!}x^4 + \frac{1}{3!}x^6 + \ldots\right] = c_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}.
\end{aligned}$$

## 6.3 Solutions about Ordinary Points

Suppose the linear second order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \tag{6.7}$$

is put into the standard form

$$y'' + P(x)y' + Q(x)y = 0 \tag{6.8}$$

by dividing by the leading coefficient $a_2(x)$.

**Definition 6.3.1.** *A point $x_0$ is said to be an* ***ordinary point*** *of the differential equation* (6.7) *if both $P(x)$ and $Q(x)$ are* ***analytic*** *at $x_0$. A point that is not an ordinary point is said to be a* ***singular point*** *of the equation.*

**Example 6.3.1.**

1. *Every finite value of $x$ is an ordinary point of $y'' + e^x y' + (\sin x)y = 0$. In particular we see that $x = 0$ is an ordinary point since both $e^x$ and $\sin x$ are analytic at this point.*

2. *The differential equation $xy'' + (\sin x)y =$ has an ordinary point at $x = 0$ since $Q(x) = \dfrac{\sin x}{x}$ is analytic at $x = 0$.*

3. *The differential equation $y'' + (\ln x)y = 0$ has a singular point at $x = 0$ because $Q(x) = \ln x$ is not analytic at $x = 0$.*

### 6.3.1 Existence of Power Series Solutions

If $x = x_0$ is an ordinary point of the differential equation (6.8), we can always find two linearly independent solutions in the form of power series centered at $x_0$

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^n. \tag{6.9}$$

A solution of a differential equation of the form given in (6.9) is said to be a solution *about* the ordinary point $x_0$.

To solve a linear second order equation such as (6.8) we find two sets of of coefficients $c_n$ so that we two distinct power series $y_1(x)$ and $y_2(x)$, both expanded about the same ordinary point $x_0$. We will assume that $y = \sum\limits_{n=0}^{\infty} c_n (x - x_0)^n$ and then determine the $c_n$. The general solution of the differential equation is

$$y = C_1 y_1(x) + C_2 y_2(x),$$

in fact, it can be shown that $C_1 = c_0$ and $C_2 = c_1$, where $c_0$ and $c_1$ are arbitrary.

For the sake of simplicity, we assume an ordinary point is always located at $x = 0$, since, if not, the substitution $t = x - x_0$ translates the value $x = x_0$ to $t = 0$.

**Example 6.3.2.** *Solve $y'' + xy = 0$.*

***Solution:*** *We see that $x = 0$ is an ordinary point of the equation. Substituting*

$$y = \sum_{n=0}^{\infty} c_n x^n \quad \textit{and} \quad y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

*into the differential equation gives*

$$\begin{aligned} y'' + xy &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1} \\ &= 2\cdot 1 c_2 x^0 + \underbrace{\sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1}}_{\textit{both series start with } x} \end{aligned}$$

*Letting* $k = n-2$ *in the first series and* $k = n+1$ *in the second, we have*

$$\begin{aligned} y'' + xy &= 2c_2 + \sum_{k=1}^{\infty}(k+2)(k+1)c_{k+2}x^k + \sum_{k=1}^{\infty} c_{k-1}x^k \\ &= 2c_2 + \sum_{k=1}^{\infty}[(k+2)(k+1)c_{k+2} + c_{k-1}]x^k = 0. \end{aligned}$$

*We must then have* $2c_2 = 0$*, which implies* $c_2 = 0$*, and*

$$(k+2)(k+1)c_{k+2} + c_{k-1} = 0.$$

*The last expression is the same as*

$$c_{k+2} = -\frac{c_{k-1}}{(k+2)(k+1)},\ k = 1, 2, 3, \ldots$$

*Iteration gives*

$$\begin{aligned} c_3 &= -\frac{c_0}{3\cdot 2} \\ c_4 &= -\frac{c_1}{4\cdot 3} \\ c_5 &= -\frac{c_2}{5\cdot 4} = 0 \\ c_6 &= -\frac{c_3}{6\cdot 5} = \frac{1}{6\cdot 5\cdot 3\cdot 2}c_0 \\ c_7 &= -\frac{c_4}{7\cdot 6} = \frac{1}{7\cdot 6\cdot 4\cdot 3}c_1 \\ c_8 &= -\frac{c_5}{8\cdot 7} = 0 \\ c_9 &= -\frac{c_6}{9\cdot 8} = -\frac{1}{9\cdot 8\cdot 6\cdot 5\cdot 3\cdot 2}c_0 \end{aligned}$$

*and so on. It should be apparent that both* $c_0$ *and* $c_1$ *are arbitrary. Now*

$$\begin{aligned} y &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + \cdots \\ &= c_0 + c_1x + 0 - \frac{1}{3\cdot 2}c_0x^3 - \frac{1}{4\cdot 3}c_1x^4 + 0 + \frac{1}{6\cdot 5\cdot 3\cdot 2}c_0x^6 + \frac{1}{7\cdot 6\cdot 4\cdot 3}c_1x^7 + 0 + \cdots \\ &= c_0\left[1 - \frac{1}{3\cdot 2}x^3 + \frac{1}{6\cdot 5\cdot 3\cdot 2}x^6 + \cdots\right] + c_1\left[x - \frac{1}{4\cdot 3}x^4 + \frac{1}{7\cdot 6\cdot 4\cdot 3}x^7 + \cdots\right]. \end{aligned}$$

**Example 6.3.3.** ***Class Doing :*** *Solve* $(x^2+1)y'' + xy' - y = 0$.

The differential equation above is called **Airy's equation** and is encountered in the study of diffraction of light, diffraction of radio waves around the surface of the Earth, aerodynamics, and the deflection of a uniform thin vertical column that bends under its own weight. Other common forms of Airy's equation are $y'' - xy = 0$ and $y'' + \alpha^2 xy = 0$.

Historically many interesting linear second order differential equations with variable coefficients arose form someone's study of a physical problem. The study of the solutions of these equations, and other functions related to these solutions, eventually gave rise to a separate branch of mathematics known as **special functions**. Moreover, some of these differential equations possess polynomial solutions obtained, of course, by trying to find power series centered at the ordinary point $x = 0$.

- $y'' - 2xy' + 2ny = 0$ **Hermite equation**.
- $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$ **Legendre's equation**.
- $(1 - x^2)y'' - xy' + n^2 y = 0$ **Chebyshev's equation**.

## 6.4 Solutions About Singular Points

### 6.4.1 Regular and Irregular Singular Points

Singular points are further classified as either regular or irregular.

**Definition 6.4.1.** *A singular point $x = x_0$ of equation (6.7) is said to be* ***regular singular point*** *if both $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at $x_0$. A singular point that is not regular is said to be an* ***irregular singular point*** *of the equation.*

## 6.5 Method of Frobenius

To solve a differential equation such as (6.7) about a regular singular point, we employ the following theorem due to Frobenius.

**Theorem 6.5.1.** *If $x = x_0$ is a regular singular point of equation (6.7), then there exists at least one solution of the form*

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n (x - x_0)^n = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}, \tag{6.10}$$

*where the number $r$ is a constant to be determined.*

**Example 6.5.1.** *Since $x = 0$ is a regular singular point of the differential equation*

$$3xy'' + y' - y = 0$$

*we try a solution of the form $y = \sum_{n=0}^{\infty} c_n x^{n+r}$. Now*

$$y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} \quad \text{and} \quad y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2},$$

*so that*

$$\begin{aligned}
3xy'' + y' - y &= 3\sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} - \sum_{n=0}^{\infty} c_n x^{n+r} \\
&= \sum_{n=0}^{\infty} (n+r)(3n+3r-2)c_n x^{n+r-1} - \sum_{n=0}^{\infty} c_n x^{n+r} \\
&= x^r \left[ r(3r-2)c_0 x^{-1} + \underbrace{\sum_{n=1}^{\infty} (n+r)(3n+3r-2)c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \right] \\
&= x^r \left[ r(3r-2)c_0 x^{-1} + \sum_{k=0}^{\infty} [(k+r+1)(3k+3r+1)c_{k+1} - c_k] x^k \right] = 0,
\end{aligned}$$

*which implies*

$$\begin{aligned}
r(3r-2)c_0 &= 0 \\
(k+r+1)(3k+3r+1)c_{k+1} - c_k &= 0, \quad k = 0, 1, 2, \ldots
\end{aligned}$$

*Since nothing is gained by taking $c_0 = 0$, we must then have*

$$r(3r-2) = 0$$

*and*

$$c_{k+1} = \frac{c_k}{(k+r+1)(3k+3r+1)}, \quad k = 0, 1, 2, \ldots$$

*The two values of $r$ satisfying $r(3r-2) = 0$, $r_1 = \frac{2}{3}$ and $r_2 = 0$, when substituted into the recurrence equation, gives two different recurrence relations,*

$$r_1 = \frac{2}{3}, \quad c_{k+1} = \frac{c_k}{(3k+5)(k+1)}, \quad k = 0, 1, 2, \ldots \tag{6.11}$$

*and*

$$r_2 = 0, \quad c_{k+1} = \frac{c_k}{(k+1)(3k+1)}, \quad k = 0, 1, 2, \ldots \tag{6.12}$$

*Iteration of* (6.11), *gives*

$$\begin{aligned}
c_1 &= \frac{c_0}{5\cdot 1}\\
c_2 &= \frac{c_1}{8\cdot 2} = \frac{c_0}{2!5\cdot 8}\\
c_3 &= \frac{c_2}{11\cdot 3} = \frac{c_0}{3!5\cdot 8\cdot 11}\\
c_4 &= \frac{c_3}{14\cdot 4} = \frac{c_0}{4!5\cdot 8\cdot 11\cdot 14}\\
&\vdots\\
c_n &= \frac{c_0}{n!5\cdot 8\cdot 11\cdots(3n+2)}, n = 1, 2, 3, \ldots
\end{aligned}$$

*whereas iteration of* (6.12) *yields*

$$\begin{aligned}
c_1 &= \frac{c_0}{1\cdot 1}\\
c_2 &= \frac{c_1}{2\cdot 4} = \frac{c_0}{2!1\cdot 4}\\
c_3 &= \frac{c_2}{3\cdot 7} = \frac{c_0}{3!1\cdot 4\cdot 7}\\
c_4 &= \frac{c_3}{4\cdot 10} = \frac{c_0}{4!1\cdot 4\cdot 7\cdot 10}\\
&\vdots\\
c_n &= \frac{c_0}{n!1\cdot 4\cdot 7\cdots(3n-2)}, \ n = 1, 2, 3, \ldots
\end{aligned}$$

*Thus we obtain two series solutions*

$$y_1 = c_0 x^{\frac{2}{3}}\left[1 + \sum_{n=1}^{\infty}\frac{1}{n!5\cdot 8\cdot 11\cdots(3n+2)}x^n\right]$$

*and*

$$y_2 = c_0 x^0\left[1 + \sum_{n=1}^{\infty}\frac{1}{n!1\cdot 4\cdot 7\cdots(3n-2)}x^n\right].$$

*Hence by the superposition principle*

$$y = C_1y_1(x) + C_2y_2(x) = C_1\left[x^{\frac{2}{3}} + \sum_{n=1}^{\infty}\frac{1}{n!5\cdot 8\cdot 11\cdots(3n+2)}x^{n+\frac{2}{3}}\right] + C_2\left[1 + \sum_{n=1}^{\infty}\frac{1}{n!1\cdot 4\cdot 7\cdots(3n-2)}x^n\right].$$

### 6.5.1 Indicial Equation

Equation $r(3r-2) = 0$ is called the **indicial equation** of the problem, and the values $r_1 = \frac{2}{3}$ and $r_2 = 0$ are called the **indicial roots** or **exponents**, of the singularity.

### 6.5.2 Cases of Indicial Roots

When using the method of Frobenius, we usually distinguish three cases corresponding to the nature of the indicial roots.

**CASE 1 Roots Not Differing By an Integer**

If $r_1$ and $r_2$ are distinct and do not differ by an integer, then there exist two linearly independent solutions of equation (6.7) of the form

$$\begin{aligned} y_1 &= \sum_{n=0}^{\infty} c_n x^{n+r_1}, c_0 \neq 0 \\ y_2 &= \sum_{n=0}^{\infty} b_n x^{n+r_2}, b_0 \neq 0 \end{aligned}$$

**Example 6.5.2.** ***Class Doing :*** *Solve* $2xy'' + (1+x)y' + y = 0.$

When the roots of the indicial equation differ by a positive integer, we may or may not be able to find two solutions. If not, then one solution corresponding to the smaller root contains a logarithmic term. When the indicial roots are equal, a second solution *always* contains a logarithm.

**CASE 2 Roots differing by a Positive Integer**

If $r_1 - r_2 = N$, where $N$ is a positive integer, then there exist two linearly independent solutions of (6.7) of the form

$$\begin{aligned} y_1 &= \sum_{n=0}^{\infty} c_n x^{n+r_1},\ c_0 \neq 0 \\ y_2 &= C y_1(x) \ln x + \sum_{n=0}^{\infty} b_n x^{n+r_2},\ b_0 \neq 0, \end{aligned}$$

where $C$ is a constant that could be zero.

**CASE 3 Equal Indicial Roots**

If $r_1 = r_2$, there always exist two linearly independent solutions of (6.7) of the form

$$\begin{aligned} y_1 &= \sum_{n=0}^{\infty} c_n x^{n+r_1},\ c_0 \neq 0 \\ y_2 &= y_1(x)\ln x + \sum_{n=1}^{\infty} b_n x^{n+r_1}. \end{aligned}$$

## 6.6 Problems

1. For each differential equation find two linearly independent power series solutions about the ordinary point $x = 0$.
   (i) $y'' - xy = 0$ (ii) $y'' - 2xy' + y = 0$ (iii) $(x^2 - 1)y'' + 4xy' + 2y = 0$
   (iv) $y'' - (x+1)y' - y = 0$ (v) $y'' - xy' - (x+2)y = 0$ (vi) $(x^2+1)y'' - 6y = 0$.

2. Show that the indicial roots do not differ by an integer. Use the method of Frobenius to obtain two linearly independent series solutions about the regular singular point $x_0 = 0$.
   (i) $2xy'' - y' + 2y = 0$ (ii) $3xy'' + (2-x)y' - y = 0$ (iii) $9x^2y'' + 9x^2y' + 2y = 0$
   (iv) $x(x-2)y'' + y' - 2y = 0$ (v) $x^2y'' - (x - \frac{2}{9})y = 0$ (vi) $2xy'' + 5y' + xy = 0$.

3. The regular singular points of the **hyper geometric equation**

   $$x(1-x)y'' + [\gamma - (\alpha + \beta + 1)x]y' - \alpha\beta y = 0,$$

   $\alpha, \beta, \gamma$ real constants are 0 and 1.

   (a) Show that the indicial roots are 0 and $1 - \gamma$.

   (b) Show that a solution corresponding to the indicial root 0 is given by

   $$y_1(x) = 1 + \frac{\alpha\beta}{1!\gamma}x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{2!\gamma(\gamma+1)}x^2 + \frac{\alpha(\alpha+1)(\alpha+2)\beta(\beta+1)(\beta+2)}{3!\gamma(\gamma+1)(\gamma+2)}x^3 + \cdots$$

   provided $\gamma$ is not zero or a negative integer. This solution is known as the **hypergeometric series**.

# Chapter 7

# The Laplace Transform

*Don't go around saying the world owes you a living. The world owes you nothing. It was here first.*

—Mark Twain

## 7.1 Introduction

The Laplace Transform was first used by and named after Pierre Simon Laplace[1], French mathematician and astronomer. Basically, a Laplace transform will convert a function in some domain into a function in another domain, without changing the value of the function. For example, let's take this function where a Laplace transform is used to convert a function of time to a function of frequency. Laplace transforms are invaluable for any engineer's mathematical toolbox as they make solving linear ODEs and related initial value problems, as well as systems of linear ODEs, much easier. Applications abound : electrical networks, springs, mixing problems, signal processing, and other areas of engineering and physics.

### 7.1.1 Why Do We Use Laplace Transform?

We use Laplace transform to convert equations having complex differential equations to relatively simple equations having polynomials. Since equations having polynomials are easier to solve, we employ Laplace transform to make calculations easier. A simple Laplace Transform is conducted while sending signals over any two-way communication medium (FM/AM stereo, 2-way radio sets, cellular phones). When information is sent over medium such as cellular phones, they are first

[1]PIERRE SIMON MARQUIS DE LAPLACE (1749-1827), great French mathematician, was a professor in Paris. He developed the foundation of potential theory and made important contributions to celestial mechanics, astronomy in general, special functions, and probability theory. Napoleon Bonaparte was his student for a year.

converted into time-varying wave, and then it is super-imposed on the medium. In this way, the information propagates. Now, at the receiving end, to decipher the information being sent, medium waves time functions are converted to frequency functions. This is a simple real life application of Laplace Transform.

### 7.1.2 Laplace Transform

**Definition 7.1.1.** *Let $f$ be a function defined for $t \geq 0$. Then the integral*

$$\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt \tag{7.1}$$

*is said to be the **Laplace transform** of $f$ provided the integral converges.*

When the defining integral (7.1) converges, the result is a function of $s$. We shall use lowercase letter to denote the function being transformed and the corresponding capital letter to denote its Laplace transform, for example

$$\mathscr{L}\{f(t)\} = F(s), \quad \mathscr{L}\{g(t)\} = G(s), \quad \mathscr{L}\{y(t)\} = Y(s).$$

Original functions depend on $x$ and their transforms on $s$ –keep this in mind!

The Laplace transform is called an integral transform because it transforms (changes) a function in one space to a function in another space by a process of integration that involves a kernel. The kernel or kernel function is a function of the variables in the two spaces and defines the integral transform.

**Example 7.1.1.** *Evaluate $\mathscr{L}\{1\}$.*

***Solution:***

$$\begin{aligned}\mathscr{L}\{1\} &= \int_0^\infty e^{st}(1)dt = \lim_{b\to\infty} \int_0^b e^{-st}dt \\ &= \lim_{b\to\infty} \left.\frac{-e^{-st}}{s}\right|_0^b = \lim_{b\to\infty} \frac{-e^{-sb}+1}{s} = \frac{1}{s}\end{aligned}$$

*provided $s > 0$. In other words, when $s > 0$, the exponent $-sb$ is negative and $e^{-sb} \to 0$ as $b \to \infty$. When $s < 0$, the integral is divergent.*

## 7.2 $\mathscr{L}$ is a Linear Transform

For a sum of functions we can write

$$\int_0^\infty e^{-st}[\alpha f(t) + \beta g(t)]dt = \alpha \int_0^\infty e^{-st} f(t)dt + \beta \int_0^\infty e^{-st} g(t)dt,$$

whenever both integrals converge. Hence, it follows that

$$\mathscr{L}\{\alpha f(t) + \beta g(t)\} = \alpha \mathscr{L}\{f(t)\} + \beta \mathscr{L}\{g(t)\} = \alpha F(s) + \beta G(s).$$

Because of this property, $\mathscr{L}$ is said to be a **linear transform**.

**Example 7.2.1.** *Evaluate* $\mathscr{L}\{e^{-3}\}$.

***Solution:*** *From the definition, we have*

$$\begin{aligned}
\mathscr{L}\{e^{-3t}\} &= \int_0^\infty e^{-st}e^{-3t}dt = \int_0^\infty e^{-(s+3)t}dt \\
&= \left.\frac{-(e^{-(s+3)t})}{s+3}\right|_0^\infty \\
&= \frac{1}{s+3},\ s > -3.
\end{aligned}$$

**Example 7.2.2.** *Evaluate* $\mathscr{L}\{f(t)\}$ *for*

$$f(t) = \begin{cases} 0, & 0 \le t < 3 \\ 2, & t \ge 3, \end{cases}$$

***Solution:*** *This piece-wise continuous function. Since $f$ is defined in two pieces, $\mathscr{L}\{f(t)\}$ is expressed as the sum of two integrals*

$$\begin{aligned}
\mathscr{L}\{f(t)\} &= \int_0^\infty e^{-st}f(t)dt \\
&= \int_0^3 e^{-st}(0)dt + \int_3^\infty e^{-st}(2)dt \\
&= -\left.\frac{2e^{-st}}{s}\right|_3^\infty \\
&= \frac{2e^{-3s}}{s},\ s > 0.
\end{aligned}$$

## 7.3 Transforms of Some Basic Functions

- $\mathscr{L}\{1\} = \dfrac{1}{s}$.
- $\mathscr{L}\{t^n\} = \dfrac{n!}{s^{n+1}},\ n = 1, 2, 3, \ldots$
- $\mathscr{L}\{e^{at}\} = \dfrac{1}{s-a}$.
- $\mathscr{L}\{\sin kt\} = \dfrac{k}{s^2+k^2}$.

- $\mathscr{L}\{\cos kt\} = \dfrac{s}{s^2+k^2}$.
- $\mathscr{L}\{\sinh kt\} = \dfrac{k}{s^2-k^2}$.
- $\mathscr{L}\{\cosh kt\} = \dfrac{s}{s^2-k^2}$.

## 7.4 Inverse Transform

In the proceeding section, we were concerned with the problem of transforming a function $f(t)$ into another function $F(s)$. We now turn to the problem around, namely, given $F(s)$, find the function $f(t)$ corresponding to this transform. We say $f(t)$ is the **inverse Laplace transform** of $F(s)$ and write

$$f(t) = \mathscr{L}^{-1}\{F(s)\}.$$

### 7.4.1 Some Inverse Transforms

- $1 = \mathscr{L}^{-1}\left\{\dfrac{1}{s}\right\}$.
- $t^n = \mathscr{L}^{-1}\left\{\dfrac{n!}{s^{n+1}}\right\}$, $n = 1, 2, 3, \ldots$
- $e^{at} = \mathscr{L}^{-1}\left\{\dfrac{1}{s-a}\right\}$.
- $\sin kt = \mathscr{L}^{-1}\left\{\dfrac{k}{s^2+k^2}\right\}$.

### 7.4.2 $\mathscr{L}^{-1}$ is a Linear Transform

We assume that the inverse Laplace transform is itself a linear transform, that is, for constants $\alpha$ and $\beta$,

$$\mathscr{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha\mathscr{L}^{-1}\{F(s)\} + \beta\mathscr{L}^{-1}\{G(s)\}$$

where $F$ and $G$ are the transforms of some functions $f$ and $g$.

**Example 7.4.1.** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{1}{s^5}\right\}$.

***Solution:*** *It follows*

$$\mathscr{L}^{-1}\left\{\frac{1}{s^5}\right\} = \frac{1}{4!}\mathscr{L}^{-1}\left\{\frac{4!}{s^5}\right\} = \frac{1}{24}t^4.$$

**Example 7.4.2.** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{3s+5}{s^2+7}\right\}$.

***Solution:*** *The given function of* $s$ *can be written as two expressions by means of term-wise division*

$$\frac{3s+5}{s^2+7} = \frac{3s}{s^2+7} + \frac{5}{s^2+7}.$$

*From the linearity property of the inverse Laplace transform, we then have*

$$\begin{aligned}\mathscr{L}^{-1}\left\{\frac{3s+5}{s^2+7}\right\} &= 3\mathscr{L}^{-1}\left\{\frac{s}{s^2+7}\right\} + \frac{5}{\sqrt{7}}\mathscr{L}^{-1}\left\{\frac{\sqrt{7}}{s^2+7}\right\} \\ &= 3\cos\sqrt{7}t + \frac{5}{\sqrt{7}}\sin\sqrt{7}t.\end{aligned}$$

### 7.4.3 Partial Fractions

Partial fractions play an important role in finding inverse Laplace transforms.

**Example 7.4.3.** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{s+1}{s^2(s+2)^3}\right\}$.

***Solution:*** *Assume*

$$\frac{s+1}{s^2(s+2)^3} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2} + \frac{E}{s+2)^3}$$

*so that* $A = -\dfrac{1}{16}, B = \dfrac{1}{8}, C = \dfrac{1}{16}, D = 0$ *and* $E = -\dfrac{1}{4}$. *Hence*

$$\begin{aligned}\mathscr{L}^{-1}\left\{\frac{s+1}{s^2(s+2)^3}\right\} &= \mathscr{L}^{-1}\left\{-\frac{1/16}{s} + \frac{1/8}{s^2} + \frac{1/16}{s+2} - \frac{1/4}{(s+2)^3}\right\} \\ &= -\frac{1}{16}\mathscr{L}^{-1}\left\{\frac{1}{s}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{16}\mathscr{L}^{-1}\{\frac{1}{s+2}\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2}{(s+2)^3}\right\} \\ &= -\frac{1}{16} + \frac{1}{8}t + \frac{1}{16}e^{-2t} - \frac{1}{8}t^2e^{-2t}.\end{aligned}$$

**Example 7.4.4.** ***Class Doing:*** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{3s-2}{s^3(s^2+4)}\right\}$.

## 7.5 Translation Theorems and Derivatives of a Transform

**Definition 7.5.1** (First Translation Theorem)**.** *If* $F(s) = \mathscr{L}\{f(t)\}$ *and* $a$ *is any real number, then*

$$\mathscr{L}\{e^{at}f(t)\} = F(s-a).$$

**Example 7.5.1.** *Evaluate (a)* $\mathscr{L}\{e^{5t}t^3\}$ *(b)* $\mathscr{L}\{e^{-2t}\cos 4t\}$.

***Solution:*** *(a)* $\mathscr{L}\{e^{5t}t^3\} = \mathscr{L}\{t^3\}_{s\to s-5} = \left.\dfrac{3!}{s^4}\right|_{s\to s-5} = \dfrac{6}{(s-5)^4}$.

*(b)* $\mathscr{L}\{e^{-2t}\cos 4t\} = \mathscr{L}\{\cos 4t\}_{s\to s+2} = \left.\dfrac{s}{s^2+16}\right|_{s\to s+2} = \dfrac{s+2}{(s+2)^2+16}$.

### 7.5.1 Inverse Form of the First Translation Theorem

If $f(t) = \mathscr{L}^{-1}\{F(s)\}$, then

$$\mathscr{L}^{-1}\{F(s-a)\} = \mathscr{L}^{-1}\{F(s)|_{s\to s-a}\} = e^{at}f(t).$$

**Example 7.5.2.** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2+6s+11}\right\}$.

***Solution:*** *If* $s^2+6s+11$ *had real factors, we would use partial fractions. Since this quadratic term does not factor, we complete the square.*

$$\begin{aligned}
\mathscr{L}^{-1}\left\{\frac{s}{s^2+6s+11}\right\} &= \mathscr{L}^{-1}\left\{\frac{s}{(s+3)^2+2}\right\} \\
&= \mathscr{L}^{-1}\left\{\frac{s+3-3}{(s+3)^2+2}\right\} \\
&= \mathscr{L}^{-1}\left\{\frac{s+3}{(s+3)^2+2} - \frac{3}{(s+3)^2+2}\right\} \\
&= \mathscr{L}^{-1}\left\{\frac{s+3}{(s+3)^2+2}\right\} - 3\mathscr{L}^{-1}\left\{\frac{1}{(s+3)^2+2}\right\} \\
&= \mathscr{L}^{-1}\left\{\left.\frac{s}{s^2+2}\right|_{s\to s+3}\right\} - \frac{3}{\sqrt{2}}\mathscr{L}^{-1}\left\{\left.\frac{\sqrt{2}}{s^2+2}\right|_{s\to s+3}\right\} \\
&= e^{-3t}\cos\sqrt{2}t - \frac{3}{\sqrt{2}}e^{-3t}\sin\sqrt{2}t.
\end{aligned}$$

**Example 7.5.3.** ***Class Doing:*** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)^3} + \dfrac{1}{s^2+2s-8}\right\}$.

## 7.6 Unit Step Function

In engineering one frequently encounters functions that can be either “on” or “off”. For example, an external force acting on a mechanical system or a voltage impressed on a circuit can be turned off after a period of time.

**Definition 7.6.1.** *The function* $\mathcal{U}(t-a)$ *is defined to be*

$$\mathcal{U}(t-a) = \begin{cases} 0, & 0 \le t < a \\ 1, & t \ge t. \end{cases}$$

When multiplied by another function defined for $t \ge 0$, the unit step function "turns off" a portion of the graph of the function.

**Example 7.6.1.**

$$f(t) = \sin t\mathcal{U}(t-2\pi) = \begin{cases} 0, & 0 \le t < 2\pi \\ \sin t, & t \ge 2\pi. \end{cases}$$

The unit step function can also be used to write piecewise defined functions in a compact form.

**Example 7.6.2.** *The piecewise defined function*

$$f(t) = \begin{cases} g(t), & 0 \le t < a \\ h(t), & t \ge a, \end{cases}$$

*is the same as* $f(t) = g(t) = g(t)\mathcal{U}(t-a) + h(t)\mathcal{U}(t-a)$.

**Example 7.6.3.** *A function of the type*

$$f(t) = \begin{cases} 0, & 0 \le t < a \\ g(t), & a \le t < b \\ 0, & t \ge b, \end{cases}$$

*can be written* $f(t) = g(t)[\mathcal{U}(t-a) - \mathcal{U}(t-b)]$.

### 7.6.1 Second Translation Theorem

**Definition 7.6.2.** *If* $F(s) = \mathscr{L}\{f(t)\}$ *and* $a > 0$, *then*

$$\mathscr{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s).$$

**Example 7.6.4.** *Evaluate* $\mathscr{L}\{(t-2)^3\mathcal{U}(t-2)\}$.

***Solution:*** $\mathscr{L}\{(t-2)^3\mathcal{U}(t-2)\} = e^{-2s}\dfrac{3!}{s^4} = \dfrac{6}{s^4}e^{-2s}$.

### 7.6.2 Inverse Form of the Second Translation Theorem

If $f(t) = \mathscr{L}^{-1}\{F(s)\}$, then

$$\mathscr{L}^{-1}\{e^{-as}Fs\} = f(t-a)\mathcal{U}(t-a).$$

**Example 7.6.5.** *Evaluate* $\mathscr{L}^{-1}\left\{\dfrac{e^{-\frac{\pi s}{2}}}{s^2+9}\right\}$.

***Solution:*** *We identify* $a = \dfrac{\pi}{2}$ *and* $f(t) = \mathscr{L}^{-1}\left\{\dfrac{1}{s^2+9}\right\} = \dfrac{1}{3}\sin 3t$. *Then*

$$\begin{aligned}
\mathscr{L}^{-1}\left\{\frac{e^{-\frac{\pi s}{2}}}{s^2+9}\right\} &= \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{3}{s^2+9}\right\}_{t\to t-\frac{\pi}{2}} \mathcal{U}\left(t-\frac{\pi}{2}\right) \\
&= \frac{1}{3}\sin 3\left(t-\frac{\pi}{2}\right)\mathcal{U}\left(t-\frac{\pi}{2}\right) \\
&= \frac{1}{3}\cos 3t\mathcal{U}\left(t-\frac{\pi}{2}\right).
\end{aligned}$$

## 7.7 Transforms of Derivatives, Integrals and Periodic Functions

Our goal is to use the Laplace transform to solve certain kinds of differential equations. To that end we need to evaluate quantities such as $\mathscr{L}\left\{\dfrac{dy}{dt}\right\}$ and $\mathscr{L}\left\{\dfrac{d^2y}{dt^2}\right\}$. For example, if $f'$ is continuous for $t \geq 0$, then integration by parts gives

$$\begin{aligned}
\mathscr{L}\{f'(t)\} &= \int_0^\infty e^{-st}f'(t)dt = e^{-st}f(t)\Big|_0^\infty + s\int_0^\infty e^{-st}f(t)dt \\
&= -f(0) + s\mathscr{L}\{f(t)\},
\end{aligned}$$

or

$$\mathscr{L}\{f'(t)\} = sF(s) - f(0).$$

Similarly

$$\begin{aligned}
\mathscr{L}\{f''(t)\} &= \int_0^\infty e^{-st}f''(t)dt = e^{-st}f'(t)\Big|_0^\infty + s\int_0^\infty e^{-st}f'(t)dt \\
&= -f'(0) + s\mathscr{L}\{f'(t)\} \\
&= s[sF(s) - f(0)] - f'(0)
\end{aligned}$$

or

$$\mathscr{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0).$$

**Theorem 7.7.1.** *If* $f(t), f'(t), \ldots, f^{n-1}(t)$ *are continuous, then*

$$\mathscr{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)},$$

*where* $F(s) = \mathscr{L}\{f(t)\}$.

## 7.8 Convolution

If functions $f$ and $g$ are piecewise continuous, then the **convolution** of $f$ and $g$, denoted by $f * g$, is given by the integral

$$f * g = \int_0^t f(\tau)g(t-\tau)d\tau.$$

**Example 7.8.1.** *The convolution of $f(t) = e^t$ and $g(t) = \sin t$ is*

$$e^t * \sin t = \int_0^t e^\tau \sin(t-\tau)d\tau = \frac{1}{2}(-\sin t - \cos t + e^t).$$

In can be shown that

$$f * g = g * f.$$

This means that the convolution of two functions is **commutative**.

### 7.8.1 Convolution Theorem

**Theorem 7.8.1.** *If $f(t)$ and $g(t)$ are piecewise continuous, then*

$$\mathscr{L}\{f * g\} = \mathscr{L}\{f(t)\}\mathscr{L}\{g(t)\} = F(s)G(s).$$

**Example 7.8.2.** *Evaluate* $\mathscr{L}\left\{\int_0^t e^\tau \sin(t-\tau)d\tau\right\}$.

***Solution:*** *With $f(t) = e^t$ and $g(t) = \sin t$, the convolution theorem states that the Laplace transform of the convolution of $f$ and $g$ is the product of their Laplace transforms,*

$$\mathscr{L}\left\{\int_0^t e^\tau \sin(t-\tau)d\tau\right\} = \mathscr{L}\{e^t\}\mathscr{L}\{\sin t\} = \frac{1}{s-1} \cdot \frac{1}{s^2+1} = \frac{1}{(s-1)(s^2+1)}.$$

### 7.8.2 Inverse Form of the Convolution Theorem

The convolution theorem is sometimes useful in finding the inverse Laplace transform of a product of two Laplace transforms, that is,

$$\mathscr{L}^{-1}\{F(s)G(s)\} = f * g.$$

**Example 7.8.3.** *Evaluate* $\mathscr{L}^{-1}\left\{\frac{1}{(s-1)(s+4)}\right\}$.

***Solution:*** *Identify* $F(s) = \dfrac{1}{s-1}$ *and* $G(s) = \dfrac{1}{s+4}$*, then* $\mathscr{L}^{-1}\{F(s)\} = f(t) = e^t$ *and* $\mathscr{L}^{}\{G(s)\} = g(t) = e^{-4t}$*. Hence*

$$
\begin{aligned}
\mathscr{L}^{-1}\left\{\frac{1}{(s-1)(s+4)}\right\} &= \int_0^t f(\tau)g(t-\tau)d\tau = \int_0^t e^{\tau}e^{-4(t-\tau)}d\tau \\
&= e^{-4t}\int_0^t e^{5\tau}d\tau \\
&= e^{-4t}\frac{1}{5}e^{5\tau}\bigg|_0^t \\
&= \frac{1}{5}e^t - \frac{1}{5}e^{-4t}.
\end{aligned}
$$

## 7.9 Applications

Since $\mathscr{L}\{y^{(n)}(t)\}, n > 1$, depends on $y(t)$ and its $n-1$ derivatives evaluated at $t = 0$, the Laplace transform is ideally suited to initial valued problems for linear differential equations with constant coefficients. This kind of differential equation can be reduced dto an *algebraic equation* in the transformed function $Y(s)$. To see this, consider the initial value problem

$$
\begin{gathered}
a_n\frac{d^ny}{dt^n} + a_{n-1}\frac{d^{n-1}y}{dt^{n-1}} + \cdots + a_1\frac{dy}{dt} + a_0y = g(t) \\
y(0) = y_0, y'(0) = y_1, \ldots, y^{(n-1)}(0) = y_{n-1},
\end{gathered}
$$

where $a_i, i = 0, 1, 2, \ldots, n$ and $y_0, y_1, \ldots, y_{n-1}$ are constants. By the linearity property of the Laplace transform, we can write

$$a_n\mathscr{L}\left\{\frac{d^ny}{dt^n}\right\} + a_{n-1}\mathscr{L}\left\{\frac{d^{n-1}y}{dt^{n-1}}\right\} + \cdots + a_0\mathscr{L}\{y\} = \mathscr{L}\{g(t)\}.$$

Therefore

$$a_n[s^nY(s) - s^{n-1}y(0) - \cdots - y^{(n-1)}(0)] + a_{n-1}[s^{n-1}Y(s) - s^{n-2}y(0) - \cdots y^{(n-2)}(0)] + \cdots + a_0Y(s) = G(s)$$

where $Y(s) = \mathscr{L}\{y(t)\}$ and $G(s) = \{g(t)\}$. By solving for $Y(s)$, we find $y(t)$ by determining the inverse transform

$$y(t) = \mathscr{L}^{-1}\{Y(s)\}.$$

**Example 7.9.1.** *Solve* $\dfrac{dy}{dt} - 3y = e^{2t}$*,* $y(0) = 1$*.*

***Solution:*** *We first take the transform of each member of the given differential equation,*

$$\mathscr{L}\left\{\frac{dy}{dt}\right\} - 3\mathscr{L}\{y\} = \mathscr{L}\{e^{2t}\}.$$

*We then use* $\mathscr{L}\left\{\dfrac{dy}{dt}\right\} = sY(s) - y(0) = sY(s) - 1$ *and* $\mathscr{L}\{e^{2t}\} = \dfrac{1}{s-2}$*. Solving*

$$sY(s) - 1 - 3Y(s) = \frac{1}{s-2}$$

*for* $Y(s)$ *and carrying out partial fraction decomposition, give*

$$Y(s) = \frac{s-1}{(s-2)(s-3)} = \frac{-1}{s-2} + \frac{2}{s-3},$$

*and so*

$$y(t) = -\mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} + 2\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\}$$

*. Then it follows that*

$$y(t) = -e^{2t} + 2e^{3t}.$$

**Example 7.9.2.** *Solve* $y'' - 6y' + 9y = t^2e^{3t}$, $y(0) = 2$, $y'(0) = 6$.

***Solution:*** $\mathscr{L}\{y''\} - 6\mathscr{L}\{y'\} + 9\{y\} = \mathscr{L}\{t^2e^{3t}\}$

$$s^2Y(s) - sy(0) - y'(0) - 6[sY(s) - y(0)] + 9Y(s) = \frac{2}{(s-3)^3}.$$

*Using the initial conditions and simplifying, give*

$$\begin{aligned}
(s^2 - 6s + 9)Y(s) &= 2s - 6 + \frac{2}{(s-3)^3} \\
(s-3)^2Y(s) &= 2(s-3) + \frac{2}{(s-3)^3} \\
Y(s) &= \frac{2}{s-3} + \frac{2}{(s-3)^5}
\end{aligned}$$

*Thus*

$$y(t) = 2\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} + \frac{2}{4!}\mathscr{L}^{-1}\left\{\frac{4!}{(s-3)^5}\right\}.$$

*Recall from the first translation theorem that*

$$\mathscr{L}^{-1}\left\{\left.\frac{4}{s^5}\right|_{s\to s-3}\right\} = t^4e^{3t}.$$

*Hence, we have*

$$y(t) = 2e^{3t} + \frac{1}{12}t^4e^{3t}.$$

**Example 7.9.3.** *Solve* $y'' + 4y' + 6y = 1 + e^{-t}$, $y(0) = 0$, $y'(0) = 0$.

***Solution:*** $\mathscr{L}\{y''\} + 4\mathscr{L}\{y'\} + 6\mathscr{L}\{y\} = \mathscr{L}\{1\} + \mathscr{L}\{e^{-t}\}$.

$$\begin{aligned}
s^2Y(s) - sy(0) - y'(0) + 4[sY(s) - y(0)] + 6Y(s) &= \frac{1}{s} + \frac{1}{s+1} \\
(s^2 + 4s + 6)Y(s) &= \frac{2s+1}{s(s+1)} \\
Y(s) &= \frac{2s+1}{s(s+1)(s^2+4s+6)}.
\end{aligned}$$

*The partial fraction decomposition for $Y(s)$ is*

$$Y(s) = \frac{1/6}{s} + \frac{1/3}{s+1} + \frac{-s/2 - 5/3}{s^2+4s+6}.$$

*In preparation for taking the inverse transform, we fix up $Y(s)$ in the following manner,*

$$Y(s) = \frac{1/6}{s} + \frac{1/3}{s+1} + \frac{(-1/2)(s+2) - 2/3}{(s+2)^2+2} = \frac{1/6}{s} + \frac{1/3}{s+1} - \frac{1}{2}\frac{s+2}{(s+2)^2+2} - \frac{2}{3}\frac{1}{(s+2)^2+2}.$$

*Finally, we obtain*

$$\begin{aligned} y(t) &= \frac{1}{6}\mathscr{L}^{-1}\left\{\frac{1}{s}\right\} = \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\} - \frac{2}{3\sqrt{2}}\mathscr{L}^{-1}\left\{\frac{\sqrt{2}}{(s+2)^2+2}\right\} \\ &= \frac{1}{6} + \frac{1}{3}e^{-t} - \frac{1}{2}e^{-2t}\cos\sqrt{2}t - \frac{\sqrt{2}}{3}\sin\sqrt{2}t. \end{aligned}$$

**Example 7.9.4.** *Solve $x'' + 16x = \cos 4t$, $x(0) = 0$, $x'(0) = 1$.*

***Solution:*** *This initial-value problem could describe the forced, undamped and resonant motion of mass on a spring. The mass starts with an initial velocity of 1 foot per second in the downward direction from the equilibrium position. Transforming the equation gives*

$$\begin{aligned} (s^2+16)X(s) &= 1 + \frac{3}{s^2+16} \\ X(s) &= \frac{1}{s^2+16} + \frac{s}{(s^2+16)^2}. \end{aligned}$$

*Therefore,*

$$\begin{aligned} x(t) &= \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{4}{s^2+16}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{8s}{(s^2+16)^2}\right\} \\ &= \frac{1}{4}\sin 4t + \frac{1}{8}t\sin 4t. \end{aligned}$$

**Example 7.9.5.** *Solve $x'' + 16x = f(t)$, $x(0) = 0$, $x'(0) = 1$, where*

$$f(t) = \begin{cases} \cos 4t, & 0 \le t < \pi \\ 0, & t \ge \pi. \end{cases}$$

***Solution:*** *The function $f(t)$ can be interpreted as an external force that is acting on a mechanical system for only a short period of time and then is removed. We can rewrite $f$ in terms of the unit step function as*

$$f(t) = \cos 4t - cos4t\mathcal{U}(t-\pi) = \cos 4t - \cos 4(t-\pi)\mathcal{U}(t-\pi).$$

*The second translation theorem then yields*

$$\mathscr{L}\{x''\} + 16\mathscr{L}\{x\} = \mathscr{L}\{f(t)\}.$$

$$
\begin{aligned}
s^2X(s) - sx(0) - x'(0) + 16X(s) &= \frac{s}{s^2+16} - \frac{s}{s^2+16}e^{-\pi s} \\
(s^2+16)X(s) &= 1 + \frac{s}{s^2+16} - \frac{s}{s^2+16}e^{-\pi s} \\
X(x) &= \frac{1}{s^2+16} + \frac{s}{(s^2+16)^2} - \frac{s}{(s^2+16)^2}e^{-\pi s}.
\end{aligned}
$$

*We then find*

$$
\begin{aligned}
x(t) &= \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{4}{s^2+16}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{8s}{(s^2+16)^2}\right\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{8s}{(s^2+16)}e^{-\pi s}\right\} \\
&= \frac{1}{4}\sin 4t + \frac{1}{8}t\sin 4t - \frac{1}{8}(t-\pi)\sin 4(t-\pi)\mathcal{U}(t-\pi).
\end{aligned}
$$

*The foregoing solution is the same as*

$$
x(t) = \begin{cases} \dfrac{1}{4}\sin 4t + \dfrac{1}{8}t\sin 4t, & 0 \le t < \pi \\[2ex] \dfrac{2+\pi}{8}\sin 4t, & t \ge \pi. \end{cases}
$$

## 7.10 Problems

1. Find $\mathscr{L}t$.
   (i) $f(t) = e^{t+7}$ (ii) $f(t) = t^2e^{3t}$ (iii) $f(t) = e^{-2t-5}$ (iv) $f(t) = t\cos t$ (v) $f(t) = te^{4t}$.

2. Find the inverse Laplace of the given transform.
   (i) $\dfrac{1}{s^3}$ (ii) $\dfrac{4s}{4s^2+1}$ (iii) $\dfrac{s-1}{s^2+2}$ (iv) $\dfrac{1}{s^2+s-20}$ (v) $\dfrac{s^2+1}{s(s-1)(s+1)(s-2)}$
   (vi) $\dfrac{6s+3}{(s^2+1)(s^2+4)}$ (vii) $\dfrac{1}{s^2(s^2+4)}$ (viii) $\dfrac{4}{s} + \dfrac{6}{s^5} - \dfrac{1}{s+8}$.

# Chapter 8

# Systems of First-Order Differential Equations

*Clothes make the man. Naked people have little or no influence on society.*

—Mark Twain

In this chapter, we confine our study to systems of first order differential equations

$$\begin{aligned}
\frac{dx_1}{dt} &= g_1(t, x_1, x_2, \dots, x_n)\\
\frac{dx_2}{dt} &= g_2(t, x_1, x_2, \dots, x_n)\\
&\vdots\\
\frac{dx_n}{dt} &= g_n(t, x_1, x_2, \dots, x_n).
\end{aligned}$$

The above system of $n$ equations is called an $n-th$ order system. When the right hand function of each equation is free of the independent variable $t$, the system is said to be **autonomous**.

### 8.0.1 Linear Systems

We have systems of linear first order equations

$$\begin{aligned}
\frac{dx_1}{dt} &= a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t)\\
\frac{dx_2}{dt} &= a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t)\\
&\vdots\\
\frac{dx_n}{dt} &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t).
\end{aligned}$$

We assume that the coefficients $a_{ij}$ and the functions $f_i$ are continuous on the common interval $J$. When $f_i, i = 1, 2, \ldots, n$, the linear system is said to be **homogeneous**, otherwise, it is **non-homogeneous**.

### 8.0.2 Matrix Form of a Linear System

If $\mathbf{X}(t)$, $\mathbf{A}(t)$ and $\mathbf{F}(t)$ denote the respective matrices

$$\mathbf{X}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \mathbf{A}(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & & \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix}, \quad \mathbf{F}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix},$$

or simply

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}.$$

If the system is homogeneous, its matrix form is then

$$\mathbf{X}' = \mathbf{A}\mathbf{X}.$$

**Example 8.0.1.** *If* $\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}$, *then the matrix form of the homogeneous system*

$$\begin{aligned} \frac{dx}{dt} &= 3x + 4y \\ \frac{dy}{dt} &= 5x - 7y \end{aligned}$$

*is*

$$\mathbf{X}' = \begin{pmatrix} 3 & 4 \\ 5 & -7 \end{pmatrix} \mathbf{X},$$

*which is an* ***autonomous system****.*

**Example 8.0.2.** *If* $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, *then the matrix form of the non-homogeneous system*

$$\begin{aligned} \frac{dx}{dt} &= 6x + y + z + t \\ \frac{dy}{dt} &= 8x + 7y - z + 10t \\ \frac{dz}{dt} &= 2x + 9y - z + 6t \end{aligned}$$

*is*

$$\mathbf{X}' = \begin{pmatrix} 6 & 1 & 1 \\ 8 & 7 & -1 \\ 2 & 9 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} t \\ 10t \\ 6t \end{pmatrix}.$$

### 8.0.3 Solution Vector

**Definition 8.0.1.** *A* ***solution vector*** *on an interval $J$ is any column vector*

$$\mathbf{X} = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

*whose entries are differentiable functions satisfying $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}$ on the interval.*